ETAS RTA-BSW v12.7.0

RTA-Xcp Stack Reference Guide

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1 XCP

Module name	Xcp
Package	RTA-XCP
AUTOSAR Module ID	212
AUTOSAR Version	22-11

1.1 Introduction

XCP (Universal Calibration Protocol) is a protocol description, based on the ASAM standard, between a master (tool) and a slave (device). The 'X' stands for variable and interchangeable transport layer.

XCP provides the following basic features.

- Synchronous data acquisition (aka measurement)
- Synchronous data stimulation (for rapid prototyping)
- Online memory calibration (read/write access)
- Calibration data page initialization and switching

XCP supports various transport layers including CAN, FlexRay and Ethernet.

1.1.1 Supported ASAM Versions

Versions of ASAM currently supported: v1.0, v1.1, v1.2, v1.3.

1.1.2 Interaction with other modules

The Xcp module has interactions with the following:

FrIf

The FlexRay Interface is used to transmit and receive XCP PDUs via FlexRay.

CanIf

The CAN Interface is used to transmit and receive XCP PDUs via CAN.

SoAd

The SocketAdaptor is used to transmit and receive XCP PDUs via Ethernet.

RTE

The RTE is used for copying calibration parameters from ROM/FLASH to RAM.

OS

In order to be able to use the time stamped feature of XCP, an AUTOSAR OS Counter is used.

1.1.3 ConfGen Support

Module Configuration

This module is not currently supported by the RTA-BSW Configuration Generator (ConfGen) and must be manually configured.

1.2 Supported Features

1.2.1 Supported Buses

The RTA-BSW implementation of XCP currently supports the following Transport Layers:

- CAN (including CAN-FD, J1939)
- FlexRay
- Ethernet (TCP and UDP)

1.2.2 Supported ASAM Features/Commands

The Xcp module supports all the basic standard commands (e.g. CONNECT, DISCONNECT, SYNCH, SET_MTA, UPLOAD, SHORT_UPLOAD). It also supports the following ASAM features/commands:

- Resume Mode
- Seed and Key
- Build Checksum
- Calibration
- Dynamic Data Acquisition (DAQ)
- Read DAQ Information
- Dynamic Data Stimulation (STIM)
- Bypassing
- Modification of separate bits
- Page switching/copying
- Check page/segment information
- Xcp Status
- User Command
- Events (but see the Limitations section)
- ODT Optimization (but see the Limitations section)

1.3 Extensions

1.3.1 Configuration Extensions

The following configuration parameters are available in version v12.7.0 in addition to those defined in AR v22-11

1.3.1.1 Xcp/XcpA2LFile

Short Name	XcpA2LFile

Long Name	Xcp A2L File
Description	Contains parameters for A2L-file generation. Only relevant if Xcp component shall generate A2L file.
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

1.3.1.2 Xcp/XcpA2LFile/XcpCoreLoad

Short Name	XcpCoreLoad
Description	This container specifies additional CPU load parameters for one CPU core for A2L-file according to ASAM XCP Specification.
Multiplicity	0..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

1.3.1.3 Xcp/XcpA2LFile/XcpCoreLoad/XcpCoreLoadMax

Short Name	XcpCoreLoadMax
Long Name	CPU Core Load Max [0=0%, 1=100%] (optional)
Description	CORELoad limit regarding all CpuLoadConsumption* parts assigned to Core(CoreNr)
Multiplicity	0..1
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

1.3.1.4 Xcp/XcpA2LFile/XcpCoreLoad/XcpCoreNr

Short Name	XcpCoreNr
Long Name	Core Nr
Description	Specifies the CPU core reference number for which the CPU load parameter are configured
Multiplicity	1..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.5 Xcp/XcpA2LFile/XcpCoreLoadMaxTotal

Short Name	XcpCoreLoadMaxTotal
Long Name	CPU Core Load Max Total [0=0%, 1=100%] (optional)

Description	TotalCORELoad limit regarding all cores
Multiplicity	0..1
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

1.3.1.6 Xcp/XcpA2LFile/XcpCpuLoadConsumptionDaq

Short Name	XcpCpuLoadConsumptionDaq
Long Name	XcpCpuLoadConsumption Type DAQ
Description	This container specifies CPU load consumption parameters of a EventPart of type DAQ according to ASAM XCP specification for one transport layer type.
Multiplicity	0..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

1.3.1.7 Xcp/XcpA2LFile/XcpCpuLoadConsumptionDaq/XcpCoreLoad-DaqEpMax

Short Name	XcpCoreLoadDaqEpMax
Long Name	Core Load EP Max Daq[0=0%, 1=100%]
Description	CORELoadDAQ limit regarding the event part: CPU_LOAD_CONSUMPTION_DAQ
Multiplicity	0..1
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

1.3.1.8 Xcp/XcpA2LFile/XcpCpuLoadConsumptionDaq/XcpCoreLoadEpRef

Short Name	XcpCoreLoadEpRef
Long Name	Core Load Ref
Description	Reference to XcpCoreLoad. This reference gives the information of CPU core number.
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

1.3.1.9 Xcp/XcpA2LFile/XcpCpuLoadConsumptionDaq/XcpCpuLoadCon-sumptionDaqDaqFactor

Short Name	XcpCpuLoadConsumptionDaqDaqFactor

Long Name	CPU Load Consumption DAQ - DAQ Factor [s]
Description	Basic CPU load to be considered for each DAQ list.
Multiplicity	1..1
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

1.3.1.10 Xcp/XcpA2LFile/XcpCpuLoadConsumptionDaq/XcpCpuLoadConsumptionDaqOdtEntryFactor

Short Name	XcpCpuLoadConsumptionDaqOdtEntryFactor
Long Name	CPU Load Consumption DAQ - OdtEntry Factor [s]
Description	CPU load caused by the handling of one single ODT entry.
Multiplicity	1..1
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

1.3.1.11 Xcp/XcpA2LFile/XcpCpuLoadConsumptionDaq/XcpCpuLoadConsumptionDaqOdtEntrySizeFactorTable

Short Name	XcpCpuLoadConsumptionDaqOdtEntrySizeFactorTable
Description	This container specifies CPU Load Consumption parameters for copy an ODT_ENTRY.
Multiplicity	1..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

1.3.1.12 Xcp/XcpA2LFile/XcpCpuLoadConsumptionDaq/XcpCpuLoadConsumptionDaqOdtEntrySizeFactorTable/XcpCpuLoadConsumptionDaqSize

Short Name	XcpCpuLoadConsumptionDaqSize
Long Name	CPU Load Consumption DAQ - Size
Description	Parameter of Odt Entry Size Factor Table for DAQ. Integer value, unit is per Byte.
Multiplicity	1..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.13 Xcp/XcpA2LFile/XcpCpuLoadConsumptionDaq/XcpCpuLoadConsumptionDaqOdtEntrySizeFactorTable/XcpCpuLoadConsumptionDaqSizeFactor

Short Name	XcpCpuLoadConsumptionDaqSizeFactor
Long Name	CPU Load Consumption DAQ - Size Factor [s]
Description	Computing time for the copying of an ODT_ENTRY contents of the magnitude SIZE. If an ODT entry has a size which is not described by any of the table records, the next smaller size entry shall be applied. If no smaller size is defined, the next larger size shall be applied.
Multiplicity	1..1
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

1.3.1.14 Xcp/XcpA2LFile/XcpCpuLoadConsumptionDaq/XcpCpuLoadConsumptionDaqOdtFactor

Short Name	XcpCpuLoadConsumptionDaqOdtFactor
Long Name	CPU Load Consumption DAQ - Odt Factor [s]
Description	Basic CPU load to be considered for each ODT.
Multiplicity	1..1
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

1.3.1.15 Xcp/XcpA2LFile/XcpCpuLoadConsumptionDaq/XcpEventChannelRef

Short Name	XcpEventChannelRef
Long Name	Event Channel Ref
Description	This XcpCpuLoadConsumption container is used by the referenced XCP event Channel
Multiplicity	1..*
Default Value	-
Type	ECUC-REFERENCE-DEF

1.3.1.16 Xcp/XcpA2LFile/XcpCpuLoadConsumptionDaq/XcpTransportLayerRef

Short Name	XcpTransportLayerRef
Long Name	TransportLayer Ref

Description	This XcpCpuLoadConsumption container is used by the referenced XCP Transpoertlayer
Multiplicity	1..*
Default Value	-
Type	ECUC-REFERENCE-DEF

1.3.1.17 Xcp/XcpA2LFile/XcpCpuLoadConsumptionQueueDaq

Short Name	XcpCpuLoadConsumptionQueueDaq
Long Name	XcpCpuLoadConsumption Type Queue
Description	This container specifies CPU load consumption parameters of a EventPart of type QUEUE according to ASAM XCP specification for one transport layer type.
Multiplicity	0..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

1.3.1.18 Xcp/XcpA2LFile/XcpCpuLoadConsumptionQueueDaq/XcpCoreLoadEpMaxQueue

Short Name	XcpCoreLoadEpMaxQueue
Long Name	Core Load Queue EP Max [0=0%, 1=100%]
Description	CORELoadQUEUE limit regarding the event part: CPU_LOAD_CONSUMPTION_QUEUE
Multiplicity	0..1
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

1.3.1.19 Xcp/XcpA2LFile/XcpCpuLoadConsumptionQueueDaq/XcpCoreLoadEpRef

Short Name	XcpCoreLoadEpRef
Long Name	Core Load Ref
Description	Reference to XcpCoreLoad. This reference gives the information of CPU core number.
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

1.3.1.20 Xcp/XcpA2LFile/XcpCpuLoadConsumptionQueueDaq/XcpCpuLoadConsumptionQueueOdtElementLoad

Short Name	XcpCpuLoadConsumptionQueueOdtElementLoad
Long Name	CPU Load Consumption Queue - OdtElement Load [s]
Description	CPU load caused by copying one single element.
Multiplicity	1..1
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

1.3.1.21 Xcp/XcpA2LFile/XcpCpuLoadConsumptionQueueDaq/XcpCpuLoadConsumptionQueueOdtFactor

Short Name	XcpCpuLoadConsumptionQueueOdtFactor
Long Name	CPU Load Consumption Queue - Odt Factor [s]
Description	Basic CPU load to be considered for buffering each ODT into the transmission queue.
Multiplicity	1..1
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

1.3.1.22 Xcp/XcpA2LFile/XcpCpuLoadConsumptionQueueDaq/XcpEventChannelRef

Short Name	XcpEventChannelRef
Long Name	Event Channel Ref
Description	If this reference is given, then this XcpCpuLoadConsumption container will be used by the referenced XCP event Channel
Multiplicity	1..*
Default Value	-
Type	ECUC-REFERENCE-DEF

1.3.1.23 Xcp/XcpA2LFile/XcpCpuLoadConsumptionQueueDaq/XcpTransportLayerRef

Short Name	XcpTransportLayerRef
Long Name	TransportLayer Ref
Description	If this reference is given, then this XcpCpuLoadConsumption container will be used by the referenced XCP Transpoertlayer
Multiplicity	1..*
Default Value	-

Type	ECUC-REFERENCE-DEF

1.3.1.24 Xcp/XcpA2LFile/XcpCpuLoadConsumptionQueueStim

Short Name	XcpCpuLoadConsumptionQueueStim
Long Name	XcpCpuLoadConsumption Type Queue STIM
Description	This container specifies CPU load consumption parameters of a EventPart of type QUEUE STIM according to ASAM XCP specification for one transport layer type.
Multiplicity	0..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

1.3.1.25 Xcp/XcpA2LFile/XcpCpuLoadConsumptionQueueStim/XcpCoreLoadEpMaxQueueStim

Short Name	XcpCoreLoadEpMaxQueueStim
Long Name	Core Load Queue EP Max STIM[0=0%, 1=100%]
Description	CORELoadQUEUE limit regarding the event part: CPU_LOAD_CONSUMPTION_QUEUE_STIM
Multiplicity	0..1
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

1.3.1.26 Xcp/XcpA2LFile/XcpCpuLoadConsumptionQueueStim/XcpCoreLoadEpRef

Short Name	XcpCoreLoadEpRef
Long Name	Core Load Ref
Description	Reference to XcpCoreLoad. This reference gives the information of CPU core number.
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

1.3.1.27 Xcp/XcpA2LFile/XcpCpuLoadConsumptionQueueStim/XcpCpuLoadConsumptionQueueStimOdtElementLoad

Short Name	XcpCpuLoadConsumptionQueueStimOdtElementLoad
Long Name	CPU Load Consumption Queue STIM - OdtElement Load [s]
Description	CPU load caused by copying one single element.

Multiplicity	1..1
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

1.3.1.28 Xcp/XcpA2LFile/XcpCpuLoadConsumptionQueueStim/XcpCpu-LoadConsumptionQueueStimOdtFactor

Short Name	XcpCpuLoadConsumptionQueueStimOdtFactor
Long Name	CPU Load Consumption Queue STIM - Odt Factor [s]
Description	Basic CPU load to be considered for buffering each ODT into the reception queue.
Multiplicity	1..1
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

1.3.1.29 Xcp/XcpA2LFile/XcpCpuLoadConsumptionQueueStim/ XcpEventChannelRef

Short Name	XcpEventChannelRef
Long Name	Event Channel Ref
Description	If this reference is given, then this XcpCpuLoadConsumption container will be used by the referenced XCP event Channel
Multiplicity	1..*
Default Value	-
Type	ECUC-REFERENCE-DEF

1.3.1.30 Xcp/XcpA2LFile/XcpCpuLoadConsumptionQueueStim/XcpTrans-portLayerRef

Short Name	XcpTransportLayerRef
Long Name	TransportLayer Ref
Description	If this reference is given, then this XcpCpuLoadConsumption container will be used by the referenced XCP Transpoertlayer
Multiplicity	1..*
Default Value	-
Type	ECUC-REFERENCE-DEF

1.3.1.31 Xcp/XcpA2LFile/XcpCpuLoadConsumptionStim

Short Name	XcpCpuLoadConsumptionStim
Long Name	XcpCpuLoadConsumption Type STIM

Description	This container specifies CPU load consumption parameters of a EventPart of type STIM according to ASAM XCP specification for one transport layer type.
Multiplicity	0..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

1.3.1.32 Xcp/XcpA2LFile/XcpCpuLoadConsumptionStim/XcpCoreLoadEpRef

Short Name	XcpCoreLoadEpRef
Long Name	Core Load Ref
Description	Reference to XcpCoreLoad. This reference gives the information of CPU core number.
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

1.3.1.33 Xcp/XcpA2LFile/XcpCpuLoadConsumptionStim/XcpCoreLoadStimEpMax

Short Name	XcpCoreLoadStimEpMax
Long Name	Core Load EP Max STIM[0=0%, 1=100%]
Description	CORELoadDAQ limit regarding the event part: CPU_LOAD_CONSUMPTION_STIM
Multiplicity	0..1
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

1.3.1.34 Xcp/XcpA2LFile/XcpCpuLoadConsumptionStim/XcpCpuLoadConsumptionStimDaqFactor

Short Name	XcpCpuLoadConsumptionStimDaqFactor
Long Name	CPU Load Consumption STIM - DAQ Factor [s]
Description	Basic CPU load to be considered for each DAQ list of the copying processes from Queue to Variable.
Multiplicity	1..1
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

1.3.1.35 Xcp/XcpA2LFile/XcpCpuLoadConsumptionStim/XcpCpuLoadConsumptionStimOdtEntryFactor

Short Name	XcpCpuLoadConsumptionStimOdtEntryFactor
Long Name	CPU Load Consumption STIM - OdtEntry Factor [s]
Description	CPU load caused by the handling of one single ODT entry of the copying processes from Queue to Variable.
Multiplicity	1..1
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

1.3.1.36 Xcp/XcpA2LFile/XcpCpuLoadConsumptionStim/XcpCpuLoadConsumptionStimOdtEntrySizeFactorTable

Short Name	XcpCpuLoadConsumptionStimOdtEntrySizeFactorTable
Description	This container specifies CPU Load Consumption parameters for copy an ODT_ENTRY.
Multiplicity	1..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

1.3.1.37 Xcp/XcpA2LFile/XcpCpuLoadConsumptionStim/XcpCpuLoadConsumptionStimOdtEntrySizeFactorTable/XcpCpuLoadConsumptionStimSize

Short Name	XcpCpuLoadConsumptionStimSize
Long Name	CPU Load Consumption STIM - Size
Description	Parameter of Odt Entry Size Factor Table for STIM. Integer value, unit is per Byte
Multiplicity	1..1
Value Range	0..65535
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.38 Xcp/XcpA2LFile/XcpCpuLoadConsumptionStim/XcpCpuLoadConsumptionStimOdtEntrySizeFactorTable/XcpCpuLoadConsumptionStimSizeFactor

Short Name	XcpCpuLoadConsumptionStimSizeFactor
Long Name	CPU Load Consumption STIM - Size Factor [s]

Description	Computing time for the copying of an ODT_ENTRY contents of the magnitude SIZE. If an ODT entry has a size which is not described by any of the table records, the next smaller size entry shall be applied. If no smaller size is defined, the next larger size shall be applied.
Multiplicity	1..1
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

1.3.1.39 Xcp/XcpA2LFile/XcpCpuLoadConsumptionStim/XcpCpuLoadConsumptionStimOdtFactor

Short Name	XcpCpuLoadConsumptionStimOdtFactor
Long Name	CPU Load Consumption STIM - Odt Factor [s]
Description	Basic CPU load to be considered for processing each ODT of the copying processes from Queue to Variable.
Multiplicity	1..1
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

1.3.1.40 Xcp/XcpA2LFile/XcpCpuLoadConsumptionStim/XcpEventChannelRef

Short Name	XcpEventChannelRef
Long Name	Event Channel Ref
Description	This XcpCpuLoadConsumption container is used by the referenced XCP event Channel
Multiplicity	1..*
Default Value	-
Type	ECUC-REFERENCE-DEF

1.3.1.41 Xcp/XcpA2LFile/XcpCpuLoadConsumptionStim/XcpTransportLayerRef

Short Name	XcpTransportLayerRef
Long Name	TransportLayer Ref
Description	This XcpCpuLoadConsumption container is used by the referenced XCP Transpoertlayer
Multiplicity	1..*
Default Value	-
Type	ECUC-REFERENCE-DEF

1.3.1.42 Xcp/XcpA2LFile/XcpCpuLoadMaxTotal

Short Name	XcpCpuLoadMaxTotal
Long Name	CPU Load Max Total [0=0%, 1=100%] (optional)
Description	Total CPU load limit regarding the DAQ measurement.
Multiplicity	0..1
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

1.3.1.43 Xcp/XcpA2LFile/XcpEventChannelA2L

Short Name	XcpEventChannelA2L
Description	This container specifies additional XcpEventChannel information for A2L-file. It can refer to a specific event channel, to a set of several event channels or to no specific event channel. When no reference is given, this event channel A2L information will be printed to all event channels, that do not have a referencing container.
Multiplicity	0..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

1.3.1.44 Xcp/XcpA2LFile/XcpEventChannelA2L/XcpCpuLoadMax

Short Name	XcpCpuLoadMax
Long Name	CPU Load Max [0=0%, 1=100%] (optional)
Description	CPU load limit for a single event.
Multiplicity	0..1
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

1.3.1.45 Xcp/XcpA2LFile/XcpEventChannelA2L/XcpEventChannelMinCycle-TimeCycle

Short Name	XcpEventChannelMinCycleTimeCycle
Long Name	Event Channel Min Cycle Time (optional)
Description	Minimum cycle time (time distance) for triggering of the event
Multiplicity	0..1
Value Range	1..255
Default Value	-

Type	ECUC-INTEGER-PARAM-DEF

1.3.1.46 Xcp/XcpA2LFile/XcpEventChannelA2L/XcpEventChannelRef

Short Name	XcpEventChannelRef
Long Name	Event Channel Ref (optional)
Description	Reference to XcpEventChannel.
Multiplicity	0..*
Default Value	-
Type	ECUC-REFERENCE-DEF

1.3.1.47 Xcp/XcpA2LFile/XcpTimeoutT1

Short Name	XcpTimeoutT1
Long Name	Timeout T1 [ms] (optional)
Description	Timeout value for slave response after command.
Multiplicity	0..1
Value Range	0..65535
Default Value	250
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.48 Xcp/XcpA2LFile/XcpTimeoutT2

Short Name	XcpTimeoutT2
Long Name	Timeout T2 [ms] (optional)
Description	Timeout value for slave response after command BUILD_CHECKSUM.
Multiplicity	0..1
Value Range	0..65535
Default Value	250
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.49 Xcp/XcpA2LFile/XcpTimeoutT7

Short Name	XcpTimeoutT7
Long Name	Timeout T7 [ms] (optional)
Description	Timeout value for master after slave error response.
Multiplicity	0..1
Value Range	0..65535
Default Value	0
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.50 Xcp/XcpA2LFile/XcpTransportLayerA2L

Short Name	XcpTransportLayerA2L
Description	This container specifies additional XcpTransportLayer information for A2L-file. These containers link the XcpEventChannelA2L information to the different transport layers.
Multiplicity	0..*
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

1.3.1.51 Xcp/XcpA2LFile/XcpTransportLayerA2L/XcpEventChannelA2LRef

Short Name	XcpEventChannelA2LRef
Long Name	Event Channel A2L Ref (optional)
Description	Reference to the XCP event channel A2L.
Multiplicity	0..*
Default Value	-
Type	ECUC-REFERENCE-DEF

1.3.1.52 Xcp/XcpA2LFile/XcpTransportLayerA2L/XcpMaxBusLoad

Short Name	XcpMaxBusLoad
Long Name	Max Bus Load [%] (optional)
Description	This indicates the maximum available busload in percent.[0 - 100%]
Multiplicity	0..1
Value Range	0..100
Default Value	100
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.53 Xcp/XcpA2LFile/XcpTransportLayerA2L/XcpOneTlPerCanCtrlBDR-Config

Short Name	XcpOneTlPerCanCtrlBDRConfig
Long Name	Create one TL for each CanControllerBaudrateConfig into the A2L-file. (optional)
Description	Enables/disables the output of one TL per CanControllerBaudrateConfig in the A2L-file. If disabled only the DefaultCanControllerBaudrateConfig is generated.
Multiplicity	0..1

Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.54 Xcp/XcpA2LFile/XcpTransportLayerA2L/XcpTransportLayerRef

Short Name	XcpTransportLayerRef
Long Name	Transport Layer Ref
Description	Reference to the XCP transport layer.
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

1.3.1.55 Xcp/XcpConfig/XcpCommunicationChannel/XcpConsistency

Short Name	XcpConsistency
Long Name	Consistency (optional)
Description	This parameter defines which CONSISTENCY is used. It has no impact on the Xcp-Implementation. The integrator has to place the calls of the EventChannel-Functions in the correct locations to achieve the requested consistency.
Multiplicity	0..1
Default Value	ODT
Enumeration Literals	EVENT, DAQ, ODT, NONE
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.56 Xcp/XcpConfig/XcpCommunicationChannel/XcpDaqMemoryLimit

Short Name	XcpDaqMemoryLimit
Long Name	DAQ memory limit [byte] (optional)
Description	The size of RAM allocated for dynamic DAQ configuration in Bytes. Xcp DAQ memory limit [byte] shall have the RAM size value which can be divided by 4. If not configured, value in XcpGeneral will be used to calculate this value. Set to zero, if you have PD RAM available, but do not want to use it with this transport layer.
Multiplicity	0..1
Value Range	0..4294967295
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.57 Xcp/XcpConfig/XcpCommunicationChannel/XcpDaqMemoryLimitEmulationDevice

Short Name	XcpDaqMemoryLimitEmulationDevice
Long Name	DAQ memory limit Emulation Device [byte] (optional, MDG1)
Description	The size of RAM allocated for dynamic DAQ configuration in Bytes in emulation device (ED) RAM area. If not configured, value in XcpGeneral will be used to calculate this value. Set to zero, if you have ED RAM available, but do not want to use it with this transport layer. Not available in Production Devices, only in Emulation Device If a ED memory is configured, Transport Layer will be duplicated in the A2L. Xcp DAQ memory limit Emulation Device [byte] shall have the RAM size value which can be divided by 4.
Multiplicity	0..1
Value Range	0..4294967295
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.58 Xcp/XcpConfig/XcpCommunicationChannel/XcpDaqOverloadIndication

Short Name	XcpDaqOverloadIndication
Long Name	DAQ Overload Indication (optional)
Description	This indicates the method used to indicate the DAQ overload situation. - NO_OVERLOAD_INDICATION indicates No overload indication - OVERLOAD_INDICATION_PID indicates overload indication in MSB of PID - OVERLOAD_INDICATION_EVENT indicates overload indication by Event Packet
Multiplicity	0..1
Default Value	NO_OVERLOAD_INDICATION
Enumeration Literals	NO_OVERLOAD_INDICATION, OVERLOAD_INDICATION_PID, OVERLOAD_INDICATION_EVENT
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.59 Xcp/XcpConfig/XcpCommunicationChannel/XcpEthernetPacketAlignment

Short Name	XcpEthernetPacketAlignment
Long Name	Ethernet Packet Alignment (optional, only Ethernet)

Description	Packet Alignment indicates the alignment a slave uses for its XCP packets. The performance for accessing the contents of the XCP messages can be increased when an XCP packet starts on an aligned address. If there is a concatenation of multiple XCP messages in one Ethernet frame, the aligning is performed by optional FILL bytes within the tail. XcpEthernetPacketAlignment shall not be configured for xcpVersion < =1.3.
Multiplicity	0..1
Default Value	-
Enumeration Literals	PACKET_ALIGNMENT_8, PACKET_ALIGNMENT_16, PACKET_ALIGNMENT_32
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.60 Xcp/XcpConfig/XcpCommunicationChannel/XcpFrameConcatenation

Short Name	XcpFrameConcatenation
Long Name	Frame Concatenation (only Ethernet and FlexRay)
Description	Enables/Disables the concatenation of several XCP protocol layer packets into one transport layer frame (with transport layer headers).
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.61 Xcp/XcpConfig/XcpCommunicationChannel/XcpOdtOptimization

Short Name	XcpOdtOptimization
Long Name	ODT Optimization (optional)
Description	This parameter defines if optimized copy-routines for copying the OdtEntries are used. It defines the largest copy-routine available. Note: This parameter does not define the largest possible datatype. A multiple of the copy-routine is also possible - OM_DEFAULT indicates no optimization. The default safe-copy routine is used - OM_ODT_TYPE_32 indicates that optimized copy routines for two and four bytes are used
Multiplicity	0..1
Default Value	OM_DEFAULT
Enumeration Literals	OM_DEFAULT, OM_ODT_TYPE_32
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.62 Xcp/XcpConfig/XcpCommunicationChannel/XcpRxPdu/

XcpEventChannelRef

Short Name	XcpEventChannelRef
Long Name	Event Channel Ref (optional, A2L-file)
Description	Reference to XcpEventChannel. XcpEventChannelRef of RxPdu must be empty for XcpVersion VERSION_1_0 and VERSION_1_1. For XCPonCAN XcpEventChannelRef can be configured if XcpRxPacketType is STIM
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

1.3.1.63 Xcp/XcpConfig/XcpCommunicationChannel/XcpRxPdu/XcpRx-PacketType

Short Name	XcpRxPacketType
Long Name	Rx Packet Type (A2L-file)
Description	This indicates the packet type for which the xcp pdu is created. - CMD indicates Command packet type - CMD_STIM indicates Command and STIM packet type - STIM indicates STIM packet type
Multiplicity	1..1
Default Value	-
Enumeration Literals	CMD, CMD_STIM, STIM, VARIABLE
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.64 Xcp/XcpConfig/XcpCommunicationChannel/XcpTransmitInMain-Function

Short Name	XcpTransmitInMainFunction
Long Name	TransmitInMainFunction (only Ethernet and Flexray)
Description	Switch to select transmission of Xcp packets on ethernet/flexray via Polling/ Joblist execution or Interrupt. If set to TRUE the transmission will be processed at task level and if set to FALSE the transmission will support interrupts.
Multiplicity	0..1
Default Value	true
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.65 Xcp/XcpConfig/XcpCommunicationChannel/XcpTransportLayer-FlexRay

Short Name	XcpTransportLayerFlexRay
Description	This container contains the additional needed XCPonFlexRay parameters in case XcpTransportLayerType is FLEXRAY
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

1.3.1.66 Xcp/XcpConfig/XcpCommunicationChannel/XcpTransportLayer-FlexRay/XcpFrNax

Short Name	XcpFrNax
Long Name	NAX (Node Address for XCP)
Description	Every node in a cluster that is an XCP slave has to have a NAX. The NAX has to be unique within the FlexRay cluster.
Multiplicity	1..1
Value Range	0..255
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.67 Xcp/XcpConfig/XcpCommunicationChannel/XcpTransportLayer-FlexRay/XcpFrPacketAlignment

Short Name	XcpFrPacketAlignment
Long Name	Packet Alignment
Description	Packet Alignment indicates the alignment a slave uses for its XCP packets. The performance for accessing the contents of the XCP messages can be increased when an XCP packet starts on an aligned address.
Multiplicity	1..1
Default Value	-
Enumeration Literals	PACKET_ALIGNMENT_8, PACKET_ALIGNMENT_16, PACKET_ALIGNMENT_32
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.68 Xcp/XcpConfig/XcpCommunicationChannel/XcpTransportLayer-FlexRay/XcpFrSequenceCorrection

Short Name	XcpFrSequenceCorrection
Long Name	Sequence Correction

Description	Enables/Disables the possibility of sequence correcting. The 1-byte field CTR is added to XCP Header and allows to detect missing packets and to correct sequencing errors.
Multiplicity	1..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.69 Xcp/XcpConfig/XcpCommunicationChannel/XcpTransportLayer-FlexRay/XcpFrTimeoutT1Flx

Short Name	XcpFrTimeoutT1Flx
Long Name	Timeout T1_FLX [ms] (optional, A2L-file)
Description	Timeout value for slave response after FlexRay transport layer command FLX_ASSIGN.
Multiplicity	0..1
Value Range	0..65535
Default Value	250
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.70 Xcp/XcpConfig/XcpCommunicationChannel/XcpTransportLayerId

Short Name	XcpTransportLayerId
Long Name	Transport Layer Id (generated)
Description	ID of the transport layer. Needed for separation of different transport layer.
Multiplicity	0..1
Value Range	0..255
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.71 Xcp/XcpConfig/XcpCommunicationChannel/XcpTransportLayer-Name

Short Name	XcpTransportLayerName
Long Name	Transport Layer Name (A2L-file)
Description	Xcp Transport Layer Name.
Multiplicity	1..1
String Length Range	1..101
Default Value	-
Type	ECUC-STRING-PARAM-DEF

1.3.1.72 Xcp/XcpConfig/XcpCommunicationChannel/XcpTransportLayer-Type

Short Name	XcpTransportLayerType
Long Name	Transport Layer Type
Description	Xcp Transport Layer Type.
Multiplicity	1..1
Default Value	-
Enumeration Literals	CAN, ETHERNET_TCP, ETHERNET_UDP, FLEXRAY, CDD
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.73 Xcp/XcpConfig/XcpCommunicationChannel/XcpTxPdu/XcpEventChannelRef

Short Name	XcpEventChannelRef
Long Name	Event Channel Ref (optional)
Description	Reference to XcpEventChannel XcpEventChannelRef of TxPdu must be empty for XcpVersion VERSION_1_0 and VERSION_1_1. XcpEventChannelRef can be configured if XcpTxPacketType is DAQ for XCPOnCAN
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

1.3.1.74 Xcp/XcpConfig/XcpCommunicationChannel/XcpTxPdu/XcpTx-PacketType

Short Name	XcpTxPacketType
Long Name	Tx Packet Type
Description	This indicates the packet type for which the Xcp PDU is created.
Multiplicity	1..1
Default Value	-
Enumeration Literals	RES_ERR, EV_SERV, RES_ERR_EV_SERV, DAQ, RES_ERR_EV_SERV_DAQ, VARIABLE
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.75 Xcp/XcpConfig/XcpEventChannel/XcpDPMProperties

Short Name	XcpDPMProperties
Long Name	Daq Packed Mode Property
Description	This Container specifies the Xcp Daq Packed Mode properties

Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

1.3.1.76 Xcp/XcpConfig/XcpEventChannel/XcpDPMProperties/XcpDPM

Short Name	XcpDPM
Long Name	Daq Packed Mode
Description	With DAQ_PACKED_MODE = 1 or DAQ_PACKED_MODE = 2, the master activates the packing of multiple samples of each data element into one DTO. Note: "ELEMENT_GROUPED" modes is not supported yet. Please avoid configuring it.
Multiplicity	0..1
Default Value	EVENT_GROUPED
Enumeration Literals	ELEMENT_GROUPED, EVENT_GROUPED
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.77 Xcp/XcpConfig/XcpEventChannel/XcpDPMProperties/XcpDPMOption

Short Name	XcpDPMOption
Long Name	Daq Packed Mode Option
Description	The parameter indicates the usage of DaqPackedMode
Multiplicity	0..1
Default Value	MANDATORY
Enumeration Literals	OPTIONAL, MANDATORY
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.78 Xcp/XcpConfig/XcpEventChannel/XcpDPMProperties/XcpDPMSampleCount

Short Name	XcpDPMSampleCount
Long Name	Daq Packed Mode Sample Count
Description	The DPMSampleCount is an unsigned 16 bit integer. It is the number of samples, which are packed into one DAQ list transmission. The values 0 and 1 are not valid.
Multiplicity	0..1
Value Range	2..65535
Default Value	3
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.79 Xcp/XcpConfig/XcpEventChannel/XcpDPMProperties/XcpDPM-SampleCountAlternate

Short Name	XcpDPMSampleCountAlternate
Long Name	Daq Packed Mode Optional SampleCount
Description	This is an optional parameter to XcpDPMSampleCount
Multiplicity	0..1
Value Range	2..65535
Default Value	4
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.80 Xcp/XcpConfig/XcpEventChannel/XcpDPMProperties/XcpDPM-TimeStampMode

Short Name	XcpDPMTimeStampMode
Long Name	Daq TimeStamp Mode
Description	The DPMTimeStampMode defines the number and meaning of time-stamps in packed mode.
Multiplicity	0..1
Default Value	STS_FIRST
Enumeration Literals	STS_LAST, STS_FIRST
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.81 Xcp/XcpConfig/XcpEventChannel/XcpDaqListPropertiesPacked-Flag

Short Name	XcpDaqListPropertiesPackedFlag
Long Name	Packed Flag (optional)
Description	The PACKED flag indicates that the DAQ list supports packed data modes for a reduced overhead with high sample rate data. The actual properties for this mode are set/queried with the commands SET_DAQ_ PACKED_MODE and GET_DAQ_PACKED_MODE.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.82 Xcp/XcpConfig/XcpEventChannel/XcpDtoCtrEventChannelProperties

Short Name	XcpDtoCtrEventChannelProperties
Long Name	Dto Ctr Event Channel property (optional)
Description	this container specifies the DTO CTR properties for an event channel
Multiplicity	0..1
Default Value	-
Type	ECUC-PARAM-CONF-CONTAINER-DEF

1.3.1.83 Xcp/XcpConfig/XcpEventChannel/XcpDtoCtrEventChannelProperties/XcpDtoCtrDaqMode

Short Name	XcpDtoCtrDaqMode
Long Name	Dto Ctr Daq Mode (optional)
Description	This parameter Indicates about the DTO CTR DAQ mode for a specific event channel Note: “INSERT_STIM_COUNTER_COPY” mode is not supported yet. Please avoid configuring it.
Multiplicity	0..1
Default Value	-
Enumeration Literals	INSERT_COUNTER, INSERT_STIM_COUNTER_COPY
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.84 Xcp/XcpConfig/XcpEventChannel/XcpDtoCtrEventChannelProperties/XcpDtoCtrDaqModeFixed

Short Name	XcpDtoCtrDaqModeFixed
Long Name	Dto Ctr Daq Mode Fixed (optional)
Description	This parameter indicates if the Dto Ctr Daq mode is fixed for the event channel
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.85 Xcp/XcpConfig/XcpEventChannel/XcpDtoCtrEventChannelProperties/XcpDtoCtrStimMode

Short Name	XcpDtoCtrStimMode
Long Name	Dto Ctr Stim Mode (optional)

Description	This parameter indicates about the DTO CTR STIM mode for a specific event channel Note: “DO_NOT_CHECK_COUNTER” mode is not supported yet. Please avoid configuring it.
Multiplicity	0..1
Default Value	-
Enumeration Literals	DO_NOT_CHECK_COUNTER, CHECK_COUNTER
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.86 Xcp/XcpConfig/XcpEventChannel/XcpDtoCtrEventChannelProperties/XcpDtoCtrStimModeFixed

Short Name	XcpDtoCtrStimModeFixed
Long Name	Dto Ctr Stim Mode Fixed (optional)
Description	This parameter indicates if the Dto Ctr Stim mode is fixed for the event channel
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.87 Xcp/XcpConfig/XcpEventChannel/XcpDtoCtrEventChannelProperties/XcpRelatedEventChannelNumber

Short Name	XcpRelatedEventChannelNumber
Long Name	Related Event Channel Number
Description	The parameter indicates the event channel the current event channel is related to.
Multiplicity	1..1
Default Value	-
Type	ECUC-REFERENCE-DEF

1.3.1.88 Xcp/XcpConfig/XcpEventChannel/XcpDtoCtrEventChannelProperties/XcpRelatedEventChannelNumberFixed

Short Name	XcpRelatedEventChannelNumberFixed
Long Name	Related Event Channel Number Fixed (optional)
Description	This parameter indicates if the related event channel is fixed or can be modified
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.89 Xcp/XcpConfig/XcpEventChannel/XcpEventChannelName

Short Name	XcpEventChannelName
Long Name	Event Channel Name (A2L-file)
Description	Event Channel Name (for a2l, max. length 101)
Multiplicity	1..1
String Length Range	1..101
Default Value	-
Type	ECUC-STRING-PARAM-DEF

1.3.1.90 Xcp/XcpConfig/XcpEventChannel/XcpEventChannelRefComplementaryBypass

Short Name	XcpEventChannelRefComplementaryBypass
Long Name	Event Channel Ref Complementary Bypass (optional, A2L-file)
Description	Reference to XcpEventChannel to print COMPLEMENTARY_BYPASS_EVENT_CHANNEL_NUMBER to A2L-file. This can be used to make a combination of two event channels building a bypassing raster. A bypassing tool can use this information to display the available bypass events of an ECU. A DAQ EventChannel shall refer to the STIM EventChannel which writes data in the same cycle. A STIM EventChannel shall refer to the DAQ EventChannel which measures data in the same cycle
Multiplicity	0..1
Default Value	-
Type	ECUC-REFERENCE-DEF

1.3.1.91 Xcp/XcpConfig/XcpEventChannel/XcpEventChannelShortName

Short Name	XcpEventChannelShortName
Long Name	Event Channel Short Name
Description	Event Channel Short Name (for SW + a2l, max. length 9). Out of this parameter, XCP generates a task to be scheduled: Xcp_EventChannel_EventChannelShortName
Multiplicity	1..1
String Length Range	1..9
Default Value	-
Type	ECUC-STRING-PARAM-DEF

1.3.1.92 Xcp/XcpConfig/XcpEventChannel/XcpODTDAQBufferElementReserveEvent

Short Name	XcpODTDAQBufferElementReserveEvent
Long Name	ODT DAQ Buffer Element Reserve Event (optional)
Description	Factor in % to increase the memory to be reserved in the send queue, direction DAQ. The parameter may be considered for each EVENT to support jitter. The parameter must only be configured for XCP version > = 1.3
Multiplicity	0..1
Value Range	0..255
Default Value	0
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.93 Xcp/XcpConfig/XcpEventChannel/XcpScheduleCallEventChannel

Short Name	XcpScheduleCallEventChannel
Long Name	Schedule Call (optional)
Description	This boolean parameter can be configurable to indicate whether a call to the Scheduler is required during the DAQ list processing for a specific event channel. This call to the Scheduler is required (Schedule Call set to TRUE) if time taken for a DAQ list processing exceeds a given time period. The time period shall be fixed by the product line which requires this feature. This is done to ensure that in co-operative multitasking, high prioritized tasks will not be blocked for too long. Enabling this parameter will cause run-time monitoring of the DAQ list at the ODT level for each ODT of the DAQ list.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.94 Xcp/XcpGeneral/XcpAddressExtension

Short Name	XcpAddressExtension
Long Name	Address Extension (optional)

Description	Enables/Disables the support of the Address Extension. Enabled when external memory like RAM, EEPROM etc (unknown to Xcp) needs to be accessed. Static Address Transformation must be enabled to use address-extension. If enabled the address-extension field in the Xcp-Commands is used. Address extension is disabled if Xcp has direct access to ECU memory or no external memory is used. If disabled this field is ignored.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.95 Xcp/XcpGeneral/XcpBlockTransferDownload

Short Name	XcpBlockTransferDownload
Long Name	Block Transfer Download (optional)
Description	Enables/Disables the ASAM XCP feature MASTER_BLOCK_MODE (block transfer from master to slave)
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.96 Xcp/XcpGeneral/XcpBlockTransferUpload

Short Name	XcpBlockTransferUpload
Long Name	Block Transfer Upload (optional)
Description	Enables/Disables the ASAM XCP feature SLAVE_BLOCK_MODE (block transfer from slave to master)
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.97 Xcp/XcpGeneral/XcpBypassConsistency

Short Name	XcpBypassConsistency
Long Name	Bypass Consistency (optional)
Description	Enable/Disable the ASAM XCP feature that ensures Bypass Consistency
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.98 Xcp/XcpGeneral/XcpCalibrationCal

Short Name	XcpCalibrationCal
Long Name	Calibration (CAL) (optional)
Description	Enables/Disables the ASAM XCP feature online memory calibration (CAL)
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.99 Xcp/XcpGeneral/XcpCmdReceivedNotification

Short Name	XcpCmdReceivedNotification
Long Name	CMD Received Notification (optional)
Description	Enables/Disables call of user function XcpAppl_CmdReceived() for every received command
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.100 Xcp/XcpGeneral/XcpCommandChecksum

Short Name	XcpCommandChecksum
Long Name	Command BUILD_CHECKSUM
Description	This is a boolean parameter which indicates if checksum is supported for BUILD_CHECKSUM command. XcpCommandChecksum supports: - False -Checksum is not supported, - True - User defined algorithm, in externally calculated function. In product line MDG1 please use USER_DEFINED for checksum calculation.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.101 Xcp/XcpGeneral/XcpCommandGetCommModeInfo

Short Name	XcpCommandGetCommModeInfo
Long Name	Command GET_COMM_MODE_INFO (optional)
Description	Enables/disabled optional command GET_COMM_MODE_INFO which provides communication supported information.
Multiplicity	0..1

Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.102 Xcp/XcpGeneral/XcpCommandGetDaqListMode

Short Name	XcpCommandGetDaqListMode
Long Name	Command GET_DAQ_LIST_MODE (optional)
Description	Enables/Disables optional command GET_DAQ_LIST_MODE which returns information on the current mode of the specified DAQ list.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.103 Xcp/XcpGeneral/XcpCommandGetId

Short Name	XcpCommandGetId
Long Name	Command GET_ID (optional)
Description	Enables/Disables command GET_ID to get the Identification from slave. User function XcpAppl_GetIdSlave() is called if enabled
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.104 Xcp/XcpGeneral/XcpCommandModifyBits

Short Name	XcpCommandModifyBits
Long Name	Command MODIFY_BITS (optional)
Description	Enables/Disables command MODIFY_BITS, which allows to modify bits in an atomic way.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.105 Xcp/XcpGeneral/XcpCommandReadDaq

Short Name	XcpCommandReadDaq
Long Name	Command READ_DAQ (optional)

Description	Enables/Disables optional command READ_DAQ. The command reads data associated with one ODT entry of a DAQ list defined by the DAQ list pointer. This command has to be disabled when Static Address Transformation is enabled to avoid reading of data that is internal to the controller.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.106 Xcp/XcpGeneral/XcpCommandUserCmd

Short Name	XcpCommandUserCmd
Long Name	Command USER_CMD (optional)
Description	Enables/Disables optional User Defined command USER_CMD
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.107 Xcp/XcpGeneral/XcpCommandsGetDaqInfo

Short Name	XcpCommandsGetDaqInfo
Long Name	Commands GET_DAQ_..._INFO (optional)
Description	Enables/Disables optional commands GET_DAQ_PROCESSOR_INFO, GET_DAQ_RESOLUTION_INFO, GET_DAQ_EVENT_INFO
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.108 Xcp/XcpGeneral/XcpCommandsGetPagInfo

Short Name	XcpCommandsGetPagInfo
Long Name	Commands GET_PAG..._INFO (optional)
Description	Enables/Disables optional commands GET_PAG_PROCESSOR_INFO, GET_SEGMENT_INFO, GET_PAGE_INFO
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.109 Xcp/XcpGeneral/XcpConnectTransportLayerCallout

Short Name	XcpConnectTransportLayerCallout
Long Name	Connect TransportLayer Callout (optional)
Description	Enables/Disables call of user function XcpAppl_ConnectTransportLayer() to allow connection from this TransportLayer or not
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.110 Xcp/XcpGeneral/XcpDaqBufferElementReserve

Short Name	XcpDaqBufferElementReserve
Long Name	DAQ Buffer Element Reserve (optional)
Description	Factor in % to increase the memory to be reserved in the send queue, direction DAQ. This value is given in percentage and is limited to a range of 0 to 255. The parameter must only be configured for XCP version >= 1.3
Multiplicity	0..1
Value Range	0..255
Default Value	0
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.111 Xcp/XcpGeneral/XcpDaqBufferElementSize

Short Name	XcpDaqBufferElementSize
Long Name	DAQ Buffer Element Size (optional)
Description	Number of memory elements to be reserved in the send queue, direction DAQ. Default value is set to 2 indicating a queue size that fits 2 measurement cycles
Multiplicity	0..1
Value Range	1..65535
Default Value	2
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.112 Xcp/XcpGeneral/XcpDaqMemoryLimit

Short Name	XcpDaqMemoryLimit
Long Name	Daq Memory Limit [byte]

Description	Total Memory for DAQ configurations for ALL protocol layers. Daq Memory Limit [byte] shall have the RAM size value which can be divided by 4. This Parameter is the calculatory base for PD transport layers' DAQ Memory Limit in A2L, if not specified differently directly in the XcpCommunicationChannel config.
Multiplicity	1..1
Value Range	0..4294967295
Default Value	-
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.113 Xcp/XcpGeneral/XcpDaqMemoryLimitEmulationDevice

Short Name	XcpDaqMemoryLimitEmulationDevice
Long Name	Daq Memory Limit Emulation Device [byte] (optional, MDG1)
Description	Total Memory for DAQ configurations for ALL protocol layers in emulation device (ED) RAM area. Not available in Production Devices, only in Emulation Device. Daq Memory Limit Emulation Device [byte] shall have the RAM size value which can be divided by 4. If you enter a value here, all transport layers with an empty DAQ Memory Limit Emulation Device will be duplicated in the A2L file. This Parameter is the calculatory base for ED transport layers' DAQ Memory Limit in A2L, if not specified differently directly in the XcpCommunicationChannel config.
Multiplicity	0..1
Value Range	0..4294967295
Default Value	0
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.114 Xcp/XcpGeneral/XcpDaqPackedMode

Short Name	XcpDaqPackedMode
Long Name	Daq Packed Mode (optional)
Description	This parameter is used to enable or disable the DAQ Packed Mode feature
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.115 Xcp/XcpGeneral/XcpDaqRamShiftSize

Short Name	XcpDaqRamShiftSize
Long Name	DAQ RAM Shift Size [byte] (optional)
Description	Defines the maximum amount of bytes which is shifted during one Xcp_MainFunction call. This value must be divisible by 4.
Multiplicity	0..1
Value Range	128..32768
Default Value	1024
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.116 Xcp/XcpGeneral/XcpDaqRamShifting

Short Name	XcpDaqRamShifting
Long Name	DAQ RAM Shifting (optional)
Description	Enables/disables shifting of DAQ RAM after DAQ config is finished by the master. This is only available, when more than one protocol layer is configured. DAQ RAM Shifting enables the optimal ram utilization, but can lead to ERR_MEM_OVERFLOW, when not enough ram for both protocol layers is available.
Multiplicity	0..1
Default Value	-
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.117 Xcp/XcpGeneral/XcpDaqStimMemoryAccessByAppl

Short Name	XcpDaqStimMemoryAccessByAppl
Long Name	DAQ/STIM Memory Access By Appl (optional)
Description	Enables/Disables call of user functions XcpAppl_OdtEntryRead() and XcpAppl_OdtEntryWrite() which read/write data. If disabled, generic build in functions are used for memory access.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.118 Xcp/XcpGeneral/XcpEthMainFunctionPeriod

Short Name	XcpEthMainFunctionPeriod

Long Name	Xcp_EthMainFunctionTx() Calling period [s] (optional)
Description	The XCP on Ethernet does not require this information but the BSW scheduler, which invokes the main function, needs it in order to plan its tasks. If Xcp on ETHERNET_TCP or Xcp on ETHERNET_UDP is configured, XcpEthMainFunctionPeriod parameter must be configured. Example Period : for 1ms = 0.001
Multiplicity	0..1
Value Range	0..INF
Default Value	-
Type	ECUC-FLOAT-PARAM-DEF

1.3.1.119 Xcp/XcpGeneral/XcpEventChannelScheduling

Short Name	XcpEventChannelScheduling
Long Name	Schedule event channels in BSW via BSWMD template
Description	If set to TRUE all the configured event channels will be scheduled in the BSW via BSWMD template.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.120 Xcp/XcpGeneral/XcpMemoryReadProtection

Short Name	XcpMemoryReadProtection
Long Name	Memory Read Protection (optional)
Description	Enables/Disables the protection for reading data from memory. User function XcpAppl_MemoryReadProtection() is called if enabled, which checks if it is allowed to read from memory. This parameter is mandatory to enable in product line MDG1.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.121 Xcp/XcpGeneral/XcpMemoryWriteProtection

Short Name	XcpMemoryWriteProtection
Long Name	Memory Write Protection (optional)

Description	Enables/Disables the protection for writing data into memory. User function XcpAppl_MemoryWriteProtection() is called if enabled, which checks if it is allowed to write into memory. This parameter is mandatory to enable in product line MDG1.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.122 Xcp/XcpGeneral/XcpMtaBlockSizeAlign

Short Name	XcpMtaBlockSizeAlign
Long Name	MTA BLOCK SIZE ALIGN (optional)
Description	This parameter provides limitations of slave device for the alignment of MTA and block size for BUILD_CHECKSUM. This parameter can be configured when XcpCommandChecksum is true. Also XcpVersion VERSION_1_3 or higher must be configured.
Multiplicity	0..1
Value Range	1..65535
Default Value	1
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.123 Xcp/XcpGeneral/XcpNewSwFlashDetect

Short Name	XcpNewSwFlashDetect
Long Name	New Software Flash Detect (optional)
Description	When New Sofware Flash Detect is enabled, after update programming, the configuration of an activate resume mode measurement will be deactivated and the related NvM block will be invalidated in order to prevent the output of implausible measurement data.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.124 Xcp/XcpGeneral/XcpNoAccessHideforDAQ

Short Name	XcpNoAccessHideforDAQ
Long Name	No Access Hide for Measurement (optional)

Description	Enables/Disables the handling of No access hide functionality during synchronous data acquisition. XcpMemoryReadProtection needs to be enabled to use No Access Hide for Measurement.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.125 Xcp/XcpGeneral/XcpOdtOptimizationErrorReaction

Short Name	XcpOdtOptimizationErrorReaction
Long Name	ODT Optimization - Error Reaction (optional)
Description	Defines the reaction if an error during the DAQ-configuration is detected. An error is e.g. if the copy routine for the OdtEntry is not the same for all OdtEntries in one Odt.From ASAM v1.4 on when the value is ABORT_CONFIGURATION and ODT Optimization is other than OM_DEFAULT the OPTIMISATION_TYPE_ODT_STRICT is generated in the A2L.
Multiplicity	0..1
Default Value	FALLBACK_TO_OM_DEFAULT
Enumeration Literals	FALLBACK_TO_OM_DEFAULT, ABORT_CONFIGURATION
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.126 Xcp/XcpGeneral/XcpPackageSecumgr

Short Name	XcpPackageSecumgr
Long Name	Secumgr Package (MDG1)
Description	"Enable/Disable to choose the header file inclusion in Xcp_ProductLineMdg1.c file. Enabled: secumgr.h will be included Disabled: rba_EcuSecu.h will be included (Default)"
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.127 Xcp/XcpGeneral/XcpPadding

Short Name	XcpPadding
Long Name	Padding Feature (optional)

Description	This feature allows padding bytes to be added to Xcp response frames (Slave to Master). The default value is set to false and it is only supported for classic CAN communication channels.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.128 Xcp/XcpGeneral/XcpPaddingByte

Short Name	XcpPaddingByte
Long Name	Padding Byte (optional)
Description	The byte to be used for padding Xcp response frames must be specified. This can only be configured if XcpPadding is enabled.
Multiplicity	0..1
Value Range	0..255
Default Value	0
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.129 Xcp/XcpGeneral/XcpPageSwitchingPag

Short Name	XcpPageSwitchingPag
Long Name	Page Switching (PAG) (optional)
Description	Enables/Disables the ASAM XCP feature page switching (PAG). XcpPageSwitchingPag may only be enabled if XcpCalibrationCal is enabled.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.130 Xcp/XcpGeneral/XcpProductLine

Short Name	XcpProductLine
Long Name	Product Line
Description	Select product line for pre-filled XCP template files and validation checks.
Multiplicity	1..1
Default Value	-
Enumeration Literals	OTHER, CSMOSAR, MDG1
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.1.131 Xcp/XcpGeneral/XcpResumeMode

Short Name	XcpResumeMode
Long Name	Resume Mode (optional)
Description	Enables/Disables the ASAM XCP feature RESUME mode
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.132 Xcp/XcpGeneral/XcpSecondSlaveInstance

Short Name	XcpSecondSlaveInstance
Long Name	Second Slave Instance (optional)
Description	If two XCP Masters shall be able to connect at the same time, enable this feature.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.133 Xcp/XcpGeneral/XcpSeedAndKey

Short Name	XcpSeedAndKey
Long Name	Seed And Key (optional)
Description	Enables/Disables the XCP Seed and Key Feature. If XcpSeedAndKey is enabled, a protected resource can be UNLOCKed (using UNLOCK command) by using SEED (Using GET_SEED command) User function XcpAppl_GetSeed is called in GET_SEED to generate seed for unlock key calculation. User function XcpAppl_Unlock is called in UNLOCK to reads key or collects parts of the key, concatenate them, then checks if given key is valid.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.134 Xcp/XcpGeneral/XcpSeedAndKeyExternalFunction

Short Name	XcpSeedAndKeyExternalFunction
Long Name	Seed And Key External Function (optional, A2L-file)

Description	Name of the Seed And Key function including file extension without path. This file is used by XCP master.
Multiplicity	0..1
Default Value	-
Type	ECUC-STRING-PARAM-DEF

1.3.1.135 Xcp/XcpGeneral/XcpStaticAddressTransformation

Short Name	XcpStaticAddressTransformation
Long Name	Static address transformation (optional)
Description	Enables/Disables the call of the user fuction XcpAppl_StaticAddressTransformation. If enabled all addresses for UPLOAD/DOWNLOAD/DAQ/STIM are passed to this function where they are replaced by another address once during compilation. When enabled uses the project specific function XcpAppl_StaticAddressTransformation to transform addresses that are unknown to Xcp and replace them with another address which can be accessed by Xcp.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.136 Xcp/XcpGeneral/XcpStimQueueFactor

Short Name	XcpStimQueueFactor
Long Name	Stim Queue Factor (optional)
Description	This parameter decides, how many memory is reserved for STIM queues. E.g. 3 means, that three complete stimulation cycles fit into the queue.
Multiplicity	0..1
Value Range	1..128
Default Value	3
Type	ECUC-INTEGER-PARAM-DEF

1.3.1.137 Xcp/XcpGeneral/XcpStoreAndClearDaq

Short Name	XcpStoreAndClearDaq
Long Name	Store Daq and Clear Daq (optional)
Description	Enables/Disables the Resume mode features of Storing and Clearing the DAQ config from NvM, used in Resume mode
Multiplicity	0..1
Default Value	false

Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.138 Xcp/XcpGeneral/XcpStoreCal

Short Name	XcpStoreCal
Long Name	Store Cal (optional)
Description	Enables/Disables the Store Calibration Request feature of the Resume mode. Calibration data can be stored in memory and retained after power off
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.139 Xcp/XcpGeneral/XcpSynchronousDataAcquisitionDaq

Short Name	XcpSynchronousDataAcquisitionDaq
Long Name	Synchronous Data Acquisition (DAQ) (optional)
Description	Enables/Disables the ASAM XCP feature Synchronous Data Acquisition (DAQ).
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.140 Xcp/XcpGeneral/XcpSynchronousDataStimulationStim

Short Name	XcpSynchronousDataStimulationStim
Long Name	Synchronous Data Stimulation (STIM) (optional)
Description	Enables/Disables the ASAM XCP feature Synchronous Data Stimulation (STIM). XcpSynchronousDataStimulationStim may only be enabled if XcpSynchronousDataAcquisitionDaq is enabled.
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.141 Xcp/XcpGeneral/XcpUploadDownloadMemoryAccessByAppl

Short Name	XcpUploadDownloadMemoryAccessByAppl
Long Name	Upload/Download Memory Access By Appl (optional)

Description	Enables/Disables call of user functions XcpAppl_MemRead() and XcpAppl_MemWrite() which read/write data. If disabled, generic build in functions are used for memory access. This parameter must be enabled for MDG1 product line
Multiplicity	0..1
Default Value	false
Type	ECUC-BOOLEAN-PARAM-DEF

1.3.1.142 Xcp/XcpGeneral/XcpVersion

Short Name	XcpVersion
Long Name	XCP Version
Description	Version of XCP protocol and transport layer. If lower version numbers are configured, not all features are supported.
Multiplicity	1..1
Default Value	-
Enumeration Literals	VERSION_1_0, VERSION_1_1, VERSION_1_2, VERSION_1_3, VERSION_1_4
Type	ECUC-ENUMERATION-PARAM-DEF

1.3.2 Additional Services

The following API are available in version v12.7.0 in addition to those defined in AR v22-11

1.3.2.1 Xcp_EventChannel

Syntax	`Xcp_EventChannel(uint16 XcpEventChannelNumber_u16)`
Parameter In	`XcpEventChannelNumber_u16` Eventchannel number to process.
Returns	The status of the requested Eventchannel
Retval	`status_u8` Returns the following Values: 0x01: XCP_STATUSBIT_DAQ_STARTED: Set when DAQ is being processed. 0x02: XCP_STATUSBIT_DAQ_ERR_OVERLOAD: Set when an overload occurs. 0x10: XCP_STATUSBIT_STIM_STARTED: Set when STIM is being processed. 0x20: XCP_STATUSBIT_STIM_ERR_NO_DATA_AVAILABLE: Set when STIM does not have a complete set of data.
Retval	`numOfReceivedStimdata_u8` A count of how many complete sets of STIM data have been received. When there are several DAQ lists in this eventchannel, the lowest counter will be taken.

Description	This function looks for DAQ lists in all protocol layers that are configured for this eventchannel. Depending on the direction of the DAQ, it will call Xcp_DaqEvent or Xcp_StimEvent, collecting the status bits to return them to the caller.

1.3.2.2 Xcp_SetControlMode

Syntax	`Xcp_SetControlMode(uint8 Xcp_CtlMode_u8)`
Parameter In	`Xcp_CtlMode_u8` Control mode to enable or to disable the following resources. Bit 0: calibration/paging activated (1) or not (0). Bit 1: reserved Bit 2: DAQ lists activated (1) or not (0). Bit 3: stimulation activated (1) or not (0). Bit 4: Flash programming activated (1) or not (0). Bit 5: reserved Bit 6: reserved Bit 7: reserved
Description	Enables or disables XCP resources.

1.3.2.3 Xcp_GetControlMode

Syntax	`Xcp_GetControlMode(void)`
Description	Returns the actual control mode or enabled resource from Xcp.

1.3.2.4 Xcp_GetConnectedTransportLayerId

Syntax	`Xcp_GetConnectedTransportLayerId(uint8 ProtocolLayerId)`
Parameter In	`ProtocolLayerId` Protocol layer Id.
Description	Returns the connected Transport Layer ID to a given protocol layer.

1.3.2.5 Xcp_GetStatus

Syntax	`Xcp_GetStatus(void)`
Returns	Status of XCP
Retval	`XcpState_en` XCP_STATE_DISCONNECTED or XCP_STATE_CONNECTED
Retval	`DaqActive_b` TRUE: Daq is active, at least one daq list in direction daq is running. FALSE: Daq is inactive, no daq list in direction daq is running.
Retval	`StimActive_b` TRUE: Stim is active, at least one daq list in direction stim is running. FALSE: Stim is inactive, no daq list in direction stim is running.

Description	Returns the actual status of XCP, Daq and Stim.

1.3.2.6 Xcp_GetStateForCalSup

Syntax	`Xcp_GetStateForCalSup(void)`
Returns	Status of XCP
Retval	`XcpState_en` XCP_STATE_DISCONNECTED or XCP_STATE_CONNECTED
Retval	`DaqState_u8` 0x01 XCP_STATE_DAQ_HAS_STOPPED: DAQ transmission has been stopped (changed since last call).
Retval	`StimState_u8` 0x01 XCP_STATE_DAQ_HAS_STOPPED: STIM transmission has been stopped (changed since last call). 0x02 XCP_STATE_ALL_STIM_DATA_RECEIVED: All STIM DAQ lists have a complete data set.
Retval	`reserved_u8` Reserved for future use.
Description	Returns the actual status of XCP.

1.3.2.7 Xcp_CanTransmit

Syntax	`Xcp_CanTransmit(PduInfoType* XcpPacketPtr, uint8 XcpTransportLayerId, Xcp_PduIdType XcpTxPduId)`
Parameters In	`XcpPacketPtr` Pointer to XcpPacket (with Length and data pointer). `XcpTransportLayerId` Global Transport Layer Id. `XcpTxPduId` XcpTxPduId (relevant for CAN only).
Returns	The status of the transmit request.
Description	This function is called by the protocol layer to transmit XcpPackets.

1.3.2.8 Xcp_CanInit

Syntax	`Xcp_CanInit(uint8 XcpTransportLayerId, uint8 XcpInitStatus)`
Parameters In	`XcpTransportLayerId` Global Transport Layer Id. `XcpInitStatus` Indicates whether or not Xcp was already initialized.
Description	Initialization of the transport layer internal variables.

1.3.2.9 Xcp_CanConnect

Syntax	`Xcp_CanConnect(uint8 XcpTransportLayerId)`
Parameters In	`XcpTransportLayerId` Global Transport Layer Id.

Description	Connect command related handling for the transport layer.

1.3.2.10 Xcp_CanDisconnect

Syntax	`Xcp_CanDisconnect(uint8 XcpTransportLayerId)`
Parameters In	`XcpTransportLayerId` Global Transport Layer Id.
Returns	The result of disconnect request.
Description	Disconnects the transport layer.

1.3.2.11 Xcp_CanTransportLayerCmd

Syntax	`Xcp_CanTransportLayerCmd(const PduInfoType* XcpCmdPacketPtr, PduInfoType* XcpResPacketPtr, uint8 XcpTransportLayerId)`
Parameters In	`XcpCmdPacketPtr` Pointer to XcpPacket with received command. `XcpTransportLayerId` Global Transport Layer Id.
Parameters Out	`XcpResPacketPtr` Pointer to XcpPacket where the response shall be stored.
Description	Handles specific Transport Layer Commands.

1.3.2.12 Xcp_CanGetTxPduId

Syntax	`Xcp_CanGetTxPduId(uint8 XcpPacketId, uint16 XcpEventChannelNumber, uint8 XcpTransportLayerId)`
Parameters In	`XcpPacketId` Type of the packet (1st byte of packets to transmit). `XcpEventChannelNumber` Event channel number (relevant for CAN only). `XcpTransportLayerId` Id if Transport Layer given by Protocol layer.
Returns	XcpTxPduId.
Description	Handles specific Transport Layer Commands.

1.3.2.13 Xcp_CanResumeModeData

Syntax	`Xcp_CanResumeModeData(uint8* DaqRamPtr, const uint8* XcpDaqRamEndPtr, uint8 XcpTransportLayerId)`
Parameters In	`DaqRamPtr` Pointer to DAQ RAM address where data is saved. `XcpDaqRamEndPtr` Last DAQ RAM address for requested Protocol Layer. `XcpTransportLayerId` Id of Transport Layer given by Protocol layer.
Returns	Error code if there's not enough space to save the data.
Description	This function is called to copy transport layer needed data for Resume Mode.

1.3.2.14 Xcp_CddTransmit

Syntax	`Xcp_CddTransmit(const PduInfoType* XcpPacketPtr, uint8 XcpTransportLayerId, Xcp_PduIdType XcpTxPduId)`
Parameters In	`XcpPacketPtr` Pointer to XcpPacket (with Length and data pointer). `XcpTransportLayerId` Global Transport Layer Id.
Returns	The status of the transmit request.
Description	This function is called by the protocol layer to transmit XcpPackets.

1.3.2.15 Xcp_CddInit

Syntax	`Xcp_CddInit(uint8 XcpTransportLayerId, uint8 XcpInitStatus)`
Parameters In	`XcpTransportLayerId` Global Transport Layer Id. `XcpInitStatus` Indicates whether or not Xcp was already initialized.
Description	Initialization of the transport layer internal variables.

1.3.2.16 Xcp_CddConnect

Syntax	`Xcp_CddConnect(uint8 XcpTransportLayerId)`
Parameters In	`XcpTransportLayerId` Global Transport Layer Id.
Description	Connect command related handling for the transport layer.

1.3.2.17 Xcp_CddDisconnect

Syntax	`Xcp_CddDisconnect(uint8 XcpTransportLayerId)`
Parameters In	`XcpTransportLayerId` Global Transport Layer Id.
Returns	The result of disconnect request.
Description	Disconnects the transport layer.

1.3.2.18 Xcp_CddTransportLayerCmd

Syntax	`Xcp_CddTransportLayerCmd(const PduInfoType* XcpCmdPacketPtr, PduInfoType* XcpResPacketPtr, uint8 XcpTransportLayerId)`
Parameters In	`XcpCmdPacketPtr` Pointer to XcpPacket with received command. `XcpTransportLayerId` Global Transport Layer Id.
Parameters Out	`XcpResPacketPtr` Pointer to XcpPacket where the response shall be stored.
Description	Handles specific Transport Layer Commands.

1.3.2.19 Xcp_CddGetTxPduId

Syntax	`Xcp_CddGetTxPduId(uint8 XcpPacketId, uint16 XcpEventChannelNumber, uint8 XcpTransportLayerId)`
Parameters In	`XcpPacketId` Type of the packet (1st byte of packets to transmit). `XcpEventChannelNumber` Event channel number (relevant for CAN only). `XcpTransportLayerId` Id if Transport Layer given by Protocol layer.
Returns	XcpTxPduId.
Description	Handles specific Transport Layer Commands.

1.3.2.20 Xcp_EthMainFunctionTx

Syntax	`Xcp_EthMainFunctionTx(void)`
Description	Xcp main function for transmission on Transport Layer.

1.3.2.21 Xcp_EthTransmit

Syntax	`Xcp_EthTransmit(const PduInfoType* XcpPacketPtr, uint8 XcpTransportLayerId, Xcp_PduIdType XcpTxPduId)`
Parameters In	`XcpPacketPtr` Pointer to XcpPacket (with Length and data pointer). `XcpTransportLayerId` Global Transport Layer Id.
Returns	The status of the transmit request.
Description	This function is called by the protocol layer to transmit XcpPackets.

1.3.2.22 Xcp_EthInit

Syntax	`Xcp_EthInit(uint8 XcpTransportLayerId, uint8 XcpInitStatus)`
Parameters In	`XcpTransportLayerId` Global Transport Layer Id. `XcpInitStatus` Indicates whether or not Xcp was already initialized.
Description	Initialization of the transport layer internal variables.

1.3.2.23 Xcp_EthConnect

Syntax	`Xcp_EthConnect(uint8 XcpTransportLayerId)`
Parameters In	`XcpTransportLayerId` Global Transport Layer Id.
Description	Connect command related handling for the transport layer.

1.3.2.24 Xcp_EthDisconnect

Syntax	`Xcp_EthDisconnect(uint8 XcpTransportLayerId)`

Parameters In	`XcpTransportLayerId` Global Transport Layer Id.
Returns	The result of disconnect request.
Description	Disconnects the transport layer.

1.3.2.25 Xcp_EthTransportLayerCmd

Syntax	`Xcp_EthTransportLayerCmd(const PduInfoType* XcpCmdPacketPtr, PduInfoType* XcpResPacketPtr, uint8 XcpTransportLayerId)`
Parameters In	`XcpCmdPacketPtr` Pointer to XcpPacket with received command. `XcpTransportLayerId` Global Transport Layer Id.
Parameters Out	`XcpResPacketPtr` Pointer to XcpPacket where the response shall be stored.
Description	Handles specific Transport Layer Commands.

1.3.2.26 Xcp_EthGetTxPduId

Syntax	`Xcp_EthGetTxPduId(uint8 XcpPacketId, uint16 XcpEventChannelNumber, uint8 XcpTransportLayerId)`
Parameters In	`XcpPacketId` Type of the packet (1st byte of packets to transmit). `XcpEventChannelNumber` Event channel number (relevant for CAN only). `XcpTransportLayerId` Id if Transport Layer given by Protocol layer.
Returns	XcpTxPduId.
Description	Handles specific Transport Layer Commands.

1.3.2.27 Xcp_EthResumeModeData

Syntax	`Xcp_EthResumeModeData(uint8* DaqRamPtr, const uint8* XcpDaqRamEndPtr, uint8 XcpTransportLayerId)`
Parameters In	`DaqRamPtr` Pointer to DAQ RAM address where data is saved. `XcpDaqRamEndPtr` Last DAQ RAM address for requested Protocol Layer. `XcpTransportLayerId` Id if Transport Layer given by Protocol layer.
Returns	Error code if there's not enough space to save the data.
Description	This function is called to copy transport layer needed data for Resume Mode.

1.3.2.28 Xcp_FrMainFunctionTx

Syntax	`Xcp_FrMainFunctionTx(void)`
Description	Xcp main function for transmission on Transport Layer.

1.3.2.29 Xcp_FrTransmit

Syntax	`Xcp_FrTransmit(const PduInfoType* XcpPacketPtr, uint8 XcpTransportLayerId, Xcp_PduIdType XcpTxPduId)`
Parameters In	`XcpPacketPtr` Pointer to XcpPacket (with Length and data pointer). `XcpTransportLayerId` Global Transport Layer Id.
Returns	The status of the transmit request.
Description	This function is called by the protocol layer to transmit XcpPackets.

1.3.2.30 Xcp_FrInit

Syntax	`Xcp_FrInit(uint8 XcpTransportLayerId, uint8 XcpInitStatus)`
Parameters In	`XcpTransportLayerId` Global Transport Layer Id. `XcpInitStatus` Indicates whether or not Xcp was already initialized.
Description	Initialization of the transport layer internal variables.

1.3.2.31 Xcp_FrConnect

Syntax	`Xcp_FrConnect(uint8 XcpTransportLayerId)`
Parameters In	`XcpTransportLayerId` Global Transport Layer Id.
Description	Connect command related handling for the transport layer.

1.3.2.32 Xcp_FrDisconnect

Syntax	`Xcp_FrDisconnect(uint8 XcpTransportLayerId)`
Parameters In	`XcpTransportLayerId` Global Transport Layer Id.
Returns	The result of disconnect request.
Description	Disconnects the transport layer.

1.3.2.33 Xcp_FrTransportLayerCmd

Syntax	`Xcp_FrTransportLayerCmd(const PduInfoType* XcpCmdPacketPtr, PduInfoType* XcpResPacketPtr, uint8 XcpTransportLayerId)`
Parameters In	`XcpCmdPacketPtr` Pointer to XcpPacket with received command. `XcpTransportLayerId` Global Transport Layer Id.
Parameters Out	`XcpResPacketPtr` Pointer to XcpPacket where the response shall be stored.
Description	Handles specific Transport Layer Commands.

1.3.2.34 Xcp_FrGetTxPduId

Syntax	`Xcp_FrGetTxPduId(uint8 XcpPacketId, uint16 XcpEventChannelNumber, uint8 XcpTransportLayerId)`
Parameters In	`XcpPacketId` Type of the packet (1st byte of packets to transmit). `XcpEventChannelNumber` Event channel number (relevant for CAN only). `XcpTransportLayerId` Id if Transport Layer given by Protocol layer.
Returns	XcpTxPduId.
Description	Handles specific Transport Layer Commands.

1.3.2.35 Xcp_FrResumeModeData

Syntax	`Xcp_FrResumeModeData(uint8* DaqRamPtr, const uint8* XcpDaqRamEndPtr, uint8 XcpTransportLayerId)`
Parameters In	`DaqRamPtr` Pointer to Daq RAM address where data is saved. `XcpDaqRamEndPtr` Last DAQ RAM address for requested Protocol Layer. `XcpTransportLayerId` Id if Transport Layer given by Protocol layer.
Returns	Error code if there's not enough space to save the data.
Description	This function is called to copy transport layer needed data for Resume Mode.

1.3.2.36 Xcp_CanDaqQueRamCalc

Syntax	`Xcp_CanDaqQueRamCalc(uint8 AddOdt, boolean setQuePointers, uint8 XcpTransportLayerId)`
Parameters In	`AddOdt` Number of Odt to add for memory calculation. `setQuePointers` TRUE: Queue pointers are set, FALSE: only memory calculation. `XcpTransportLayerId` Id if Transport Layer given by Protocol layer.
Returns	Bytes required for daqlist queue.
Description	Calculates the memory consumption of the daq queue and initializes the queue pointers.

1.3.2.37 Xcp_CddDaqQueRamCalc

Syntax	`Xcp_CddDaqQueRamCalc(uint8 AddOdt, boolean setQuePointers, uint8 XcpTransportLayerId)`
Parameters In	`AddOdt` Number of Odt to add for memory calculation. `setQuePointers` TRUE: Queue pointers are set, FALSE: only memory calculation. `XcpTransportLayerId` Id if Transport Layer given by Protocol layer.
Returns	Bytes required for daqlist queue.

Description	Calculates the memory consumption of the daq queue and initializes the queue pointers.

1.3.2.38 Xcp_EthDaqQueRamCalc

Syntax	`Xcp_EthDaqQueRamCalc(uint8 AddOdt, boolean setQuePointers, uint8 XcpTransportLayerId)`
Parameters In	`AddOdt` Number of Odt to add for memory calculation. `setQuePointers` TRUE: Queue pointers are set, FALSE: only memory calculation. `XcpTransportLayerId` Id if Transport Layer given by Protocol layer.
Returns	Bytes required for daqlist queue.
Description	Calculates the memory consumption of the daq queue and initializes the queue pointers.

1.3.2.39 Xcp_FrDaqQueRamCalc

Syntax	`Xcp_FrDaqQueRamCalc(uint8 AddOdt, boolean setQuePointers, uint8 XcpTransportLayerId)`
Parameters In	`AddOdt` Number of Odt to add for memory calculation. `setQuePointers` TRUE: Queue pointers are set, FALSE: only memory calculation. `XcpTransportLayerId` Id if Transport Layer given by Protocol layer.
Returns	Bytes required for daqlist queue.
Description	Calculates the memory consumption of the daq queue and initializes the queue pointers.

1.4 Limitations

1.4.1 Usage

Xcp can read and write any part of memory under the control of an external agent, and it is generally not possible to detect and enter a safe state for all possible errors that might be caused by Xcp. For these reasons, you must ensure that Xcp is only used in an environment where failure of the ECU has acceptable consequences.

1.4.2 General Limitations

The following AUTOSAR-defined features are not currently supported in the RTA-BSW implementation of Xcp:

- Flash Programming
- Time Correlation
- Static DAQ/STIM

Unconfigured Event Channels in the command *SET_DAQ_LIST_MODE* are no longer supported due to safety concerns i.e. they can lead to out-of-bounds access.

To be able to use the Bypass feature, the component *rba_Svbb* of the package *CalSup* is necessary. (Bypass has been migrated from *rba_Byp* to *rba_Svbb*.) Xcp forwards the calculated values of the maximum number of *OdtEntries* and memory required to *rba_Svbb*.

1.4.3 Resume Mode

Resume mode is currently supported only for Protocol Layer 0. If you try to request Resume mode on a Protocol Layer other than 0, you'll receive an ERR_-GENERIC response and no data will be saved to NvM.

1.4.4 ASAM Commands

The following ASAM Features/Commands are not currently supported:

Events (EV)	Currently Not Supported: - EV_STORE_CAL - EV_CMD_PENDING - EV_SESSION_TERMINATED - EV_TIME_SYNC - EV_SLEEP - EV_WAKE_UP - EV_USER - EV_TRANSPORT
Service requests (SERV)	Currently no service requests are supported. - SERV_RESET - SERV_TEXT
Calibration data page freezing	Freeze Mode is not supported. - SET_SEGMENT_MODE - GET_SEGMENT_MODE

1.4.5 ODT Optimization

As noted in the Supported Features section, the Xcp module supports ODT Optimization, but the following limitations should be noted:

- Currently only copy routines up to 32-bit are supported.
- MAX_DTO has to be higher than 8.
- The ID field has to be relative word-aligned and the timestamp has to be

either none or four byte.

1.4.6 Generated A2L File

It is not possible to include the generated A2L file in the template A2L file (either version 1.2 or 1.3) without modification. Any attempt to include the generated A2L file in the template A2L file will throw an error when the A2L file is imported. Use the following workaround:

- Generate the A2L file during XCP CodeGen as normal.
- Copy the contents of the generated file into a new/consolidated file which also contains the content of the XCP A2L template file, the aml file (these files depend on the version of XCP), and the definition of the calibration and measurement variables.

1.5 Exclusions

The following features from AR v22-11 are not supported in version v12.7.0.

API functions not currently implemented:

- `Xcp_TransmissionModeType`
- `Xcp_SetTransmissionMode`

The following configuration parameters from AR v22-11 are not supported in version v12.7.0.

- Xcp/XcpConfig/XcpCommunicationChannel/XcpChannelRxPduRef
- Xcp/XcpConfig/XcpCommunicationChannel/XcpChannelTxPduRef
- Xcp/XcpConfig/XcpCommunicationChannel/XcpComMChannelRef
- Xcp/XcpConfig/XcpDaqList
- Xcp/XcpConfig/XcpDaqList/XcpDaqListNumber
- Xcp/XcpConfig/XcpDaqList/XcpDaqListType
- Xcp/XcpConfig/XcpDaqList/XcpDto
- Xcp/XcpConfig/XcpDaqList/XcpDto/XcpDto2PduMapping
- Xcp/XcpConfig/XcpDaqList/XcpDto/XcpDtoPid
- Xcp/XcpConfig/XcpDaqList/XcpMaxOdt
- Xcp/XcpConfig/XcpDaqList/XcpMaxOdtEntries
- Xcp/XcpConfig/XcpDaqList/XcpOdt
- Xcp/XcpConfig/XcpDaqList/XcpOdt/XcpOdt2DtoMapping
- Xcp/XcpConfig/XcpDaqList/XcpOdt/XcpOdtEntry
- Xcp/XcpConfig/XcpDaqList/XcpOdt/XcpOdtEntry/XcpOdtEntryAddress
- Xcp/XcpConfig/XcpDaqList/XcpOdt/XcpOdtEntry/XcpOdtEntryBitOffset
- Xcp/XcpConfig/XcpDaqList/XcpOdt/XcpOdtEntry/XcpOdtEntryLength

- Xcp/XcpConfig/XcpDaqList/XcpOdt/XcpOdtEntry/XcpOdtEntryNumber
- Xcp/XcpConfig/XcpDaqList/XcpOdt/XcpOdtEntryMaxSize
- Xcp/XcpConfig/XcpDaqList/XcpOdt/XcpOdtNumber
- Xcp/XcpConfig/XcpEventChannel/XcpEventChannelConsistency
- Xcp/XcpConfig/XcpEventChannel/XcpEventChannelMaxDaqList
- Xcp/XcpConfig/XcpEventChannel/XcpEventChannelTriggeredDaqListRef
- Xcp/XcpConfig/XcpPageSwitching
- Xcp/XcpConfig/XcpPageSwitching/XcpSegment
- Xcp/XcpConfig/XcpPageSwitching/XcpSegment/XcpPage
- Xcp/XcpConfig/XcpPageSwitching/XcpSegment/XcpPage/XcpPageAddress
- Xcp/XcpConfig/XcpPageSwitching/XcpSegment/XcpReferencePageAd-dress
- Xcp/XcpConfig/XcpPageSwitching/XcpSegment/XcpSegmentLength
- Xcp/XcpConfig/XcpPageSwitching/XcpSegment/XcpWorkingPageAddress
- Xcp/XcpConfig/XcpPdu
- Xcp/XcpGeneral/XcpCounterRef
- Xcp/XcpGeneral/XcpDaqConfigType
- Xcp/XcpGeneral/XcpDaqCount
- Xcp/XcpGeneral/XcpFlashProgrammingEnabled
- Xcp/XcpGeneral/XcpMaxEventChannel
- Xcp/XcpGeneral/XcpMinDaq
- Xcp/XcpGeneral/XcpOdtCount
- Xcp/XcpGeneral/XcpOdtEntriesCount
- Xcp/XcpGeneral/XcpOdtEntrySizeDaq
- Xcp/XcpGeneral/XcpOdtEntrySizeStim
- Xcp/XcpGeneral/XcpOnCanEnabled
- Xcp/XcpGeneral/XcpOnCddEnabled
- Xcp/XcpGeneral/XcpOnEthernetEnabled
- Xcp/XcpGeneral/XcpOnFlexRayEnabled
- Xcp/XcpGeneral/XcpPrescalerSupported
- Xcp/XcpGeneral/XcpSuppressTxSupport

1.6 Deviations

1.6.1 Configuration Deviations

Configuration parameters which differ from the AR v22-11 specification are listed below.

1.6.1.1 Xcp

[RTA-BSW] Description	Configuration of the XCP (Universal Calibration Protocol) module with vendor specific enhancements of RB).
[AUTOSAR] Description	Configuration of the XCP module

1.6.1.2 Xcp/XcpConfig

[RTA-BSW] Description	Xcp Configuration
[AUTOSAR] Description	This container contains the configuration parameters and sub containers of the AUTOSAR Xcp module.

1.6.1.3 Xcp/XcpConfig/XcpCommunicationChannel

[RTA-BSW] Description	This container contains the configuration of the transport layers.
[AUTOSAR] Description	This container represents the configuration of the communication channel of XCP.
[RTA-BSW] Lower Multiplicity	1
[AUTOSAR] Lower Multiplicity	0

1.6.1.4 Xcp/XcpConfig/XcpCommunicationChannel/XcpIdentification-FieldType

[RTA-BSW] Parent Container	Xcp/XcpConfig/XcpCommunicationChannel/XcpIdentificationFieldType
[AUTOSAR] Parent Container	Xcp/XcpGeneral/XcpIdentificationFieldType
[RTA-BSW] Description	Type of Identification Field the slave will use when transferring DAQ Packets to the master. The master has to use the same Type of Identification Field when transferring STIM Packets to the slave. - ABSOLUTE indicates identification type Absolute ODT number - RELATIVE_BYTE indicates identification field type Relative ODT number, absolute DAQ list number (BYTE) - RELATIVE_WORD indicates identification field type Relative ODT number, absolute DAQ list number (WORD) - RELATIVE_WORD_ALIGNED indicates identification field type Relative ODT number, absolute DAQ list number (WORD, aligned)
[AUTOSAR] Description	Type of Identification Field the slave will use when transferring DAQ Packets to the master. The master has to use the same Type of Identification Field when transferring STIM Packets to the slave.

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1
[ADDED] Default Value	ABSOLUTE

1.6.1.5 Xcp/XcpConfig/XcpCommunicationChannel/XcpMaxCto

[RTA-BSW] Parent Container	Xcp/XcpConfig/XcpCommunicationChannel/XcpMaxCto
[AUTOSAR] Parent Container	Xcp/XcpGeneral/XcpMaxCto
[RTA-BSW] Description	MAX_CTO shows the maximum length of a CTO packet in bytes for this transport layer.
[AUTOSAR] Description	MAX_CTO shows the maximum length of a CTO packet in bytes.

1.6.1.6 Xcp/XcpConfig/XcpCommunicationChannel/XcpMaxDto

[RTA-BSW] Parent Container	Xcp/XcpConfig/XcpCommunicationChannel/XcpMaxDto
[AUTOSAR] Parent Container	Xcp/XcpGeneral/XcpMaxDto
[RTA-BSW] Description	MAX_DTO shows the maximum length of a DTO packet in bytes for this transport layer.
[AUTOSAR] Description	MAX_DTO shows the maximum length of a DTO packet in bytes.

1.6.1.7 Xcp/XcpConfig/XcpCommunicationChannel/XcpRxPdu

[RTA-BSW] Parent Container	Xcp/XcpConfig/XcpCommunicationChannel/XcpRxPdu
[AUTOSAR] Parent Container	Xcp/XcpConfig/XcpPdu/XcpRxPdu
[RTA-BSW] Multiplicity	1..*
[AUTOSAR] Multiplicity	0..1

1.6.1.8 Xcp/XcpConfig/XcpCommunicationChannel/XcpRxPdu/XcpRxPduId

[RTA-BSW] Parent Container	Xcp/XcpConfig/XcpCommunicationChannel/XcpRxPdu/XcpRxPduId
[AUTOSAR] Parent Container	Xcp/XcpConfig/XcpPdu/XcpRxPdu/XcpRxPduId
[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

1.6.1.9 Xcp/XcpConfig/XcpCommunicationChannel/XcpRxPdu/XcpRxPduRef

[RTA-BSW] Parent Container	Xcp/XcpConfig/XcpCommunicationChannel/XcpRxPdu/XcpRxPduRef
[AUTOSAR] Parent Container	Xcp/XcpConfig/XcpPdu/XcpRxPdu/XcpRxPduRef
[ADDED] Description	This name uniquely identifies a group of type “XcpRxPduRef”

1.6.1.10 Xcp/XcpConfig/XcpCommunicationChannel/XcpTimestampType

[RTA-BSW] Parent Container	Xcp/XcpConfig/XcpCommunicationChannel/XcpTimestampType
[AUTOSAR] Parent Container	Xcp/XcpGeneral/XcpTimestampType
[RTA-BSW] Description	This parameter indicates the number of bytes used for the timestamp field. In case No_TIME_STAMP is selected the timestamp field is not available. - NO_TIME_STAMP indicates no timestamp field - ONE_BYTE indicates one byte timestamp field - TWO_BYTE indicates two byte timestamp field - FOUR_BYTE indicates four bytes timestamp field
[AUTOSAR] Description	This parameter indicates the number of bytes used for the timestamp field. In case No_TIME_STAMP is selected the timestamp field is not available.
[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1
[RTA-BSW] Enumeration Literals	NO_TIME_STAMP, ONE_BYTE, TWO_BYTE, FOUR_BYTE
[AUTOSAR] Enumeration Literals	FOUR_BYTE, NO_TIME_STAMP, ONE_BYTE, TWO_BYTE
[ADDED] Default Value	NO_TIME_STAMP

1.6.1.11 Xcp/XcpConfig/XcpCommunicationChannel/XcpTxPdu

[RTA-BSW] Parent Container	Xcp/XcpConfig/XcpCommunicationChannel/XcpTxPdu

[AUTOSAR] Parent Container	Xcp/XcpConfig/XcpPdu/XcpTxPdu
[RTA-BSW] Multiplicity	1..*
[AUTOSAR] Multiplicity	0..1

1.6.1.12 Xcp/XcpConfig/XcpCommunicationChannel/XcpTxPdu/XcpTxPduId

[RTA-BSW] Parent Container	Xcp/XcpConfig/XcpCommunicationChannel/XcpTxPdu/XcpTxPduId
[AUTOSAR] Parent Container	Xcp/XcpConfig/XcpPdu/XcpTxPdu/XcpTxPduId
[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

1.6.1.13 Xcp/XcpConfig/XcpCommunicationChannel/XcpTxPdu/XcpTxPduRef

[RTA-BSW] Parent Container	Xcp/XcpConfig/XcpCommunicationChannel/XcpTxPdu/XcpTxPduRef
[AUTOSAR] Parent Container	Xcp/XcpConfig/XcpPdu/XcpTxPdu/XcpTxPduRef

1.6.1.14 Xcp/XcpConfig/XcpEventChannel

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

1.6.1.15 Xcp/XcpConfig/XcpEventChannel/XcpEventChannelNumber

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1

1.6.1.16 Xcp/XcpConfig/XcpEventChannel/XcpEventChannelPriority

[RTA-BSW] Description	Priority of the event channel (0xFF = highest)
[AUTOSAR] Description	Priority of the event channel

1.6.1.17 Xcp/XcpConfig/XcpEventChannel/XcpEventChannelTimeCycle

[RTA-BSW] Description	The event channel time cycle indicates which sampling period is used to process this event channel. A value of 0 means “Not cyclic”
[AUTOSAR] Description	The event channel time cycle indicates which sampling period is used to process this event channel. A value of 0 means ‘Not cyclic’.

1.6.1.18 Xcp/XcpConfig/XcpEventChannel/XcpEventChannelTimeUnit

[RTA-BSW] Description	This configuration parameter indicates the unit of the event channel time cycle. - TIMESTAMP_UNIT_100MS indicates Unit is 100 millisecond - TIMESTAMP_UNIT_100NS indicates Unit is 100 nanosecond - TIMESTAMP_UNIT_100PS indicates Unit is 100 picosecond - TIMESTAMP_UNIT_100US indicates Unit is 100 microsecond - TIMESTAMP_UNIT_10MS indicates Unit is 10 millisecond - TIMESTAMP_UNIT_10NS indicates Unit is 10 nanosecond - TIMESTAMP_UNIT_10PS indicates Unit is 10 picosecond - TIMESTAMP_UNIT_10US indicates Unit is 10 microsecond - TIMESTAMP_UNIT_1MS indicates Unit is 1 millisecond - TIMESTAMP_UNIT_1NS indicates Unit is 1 nanosecond - TIMESTAMP_UNIT_1PS indicates Unit is 1 picosecond - TIMESTAMP_UNIT_1S indicates Unit is 1 second - TIMESTAMP_UNIT_1US indicates Unit is 1 microsecond
[AUTOSAR] Description	This configuration parameter indicates the unit of the event channel time cycle.
[RTA-BSW] Lower Multiplicity	1
[AUTOSAR] Lower Multiplicity	0
[RTA-BSW] Enumeration Literals	TIMESTAMP_UNIT_1PS, TIMESTAMP_UNIT_10PS, TIMESTAMP_UNIT_100PS, TIMESTAMP_UNIT_1NS, TIMESTAMP_UNIT_10NS, TIMESTAMP_UNIT_100NS, TIMESTAMP_UNIT_1US, TIMESTAMP_UNIT_10US, TIMESTAMP_UNIT_100US, TIMESTAMP_UNIT_1MS, TIMESTAMP_UNIT_10MS, TIMESTAMP_UNIT_100MS, TIMESTAMP_UNIT_1S
[AUTOSAR] Enumeration Literals	TIMESTAMP_UNIT_100MS, TIMESTAMP_UNIT_100NS, TIMESTAMP_UNIT_100PS, TIMESTAMP_UNIT_100US, TIMESTAMP_UNIT_10MS, TIMESTAMP_UNIT_10NS, TIMESTAMP_UNIT_10PS, TIMESTAMP_UNIT_10US, TIMESTAMP_UNIT_1MS, TIMESTAMP_UNIT_1NS, TIMESTAMP_UNIT_1PS, TIMESTAMP_UNIT_1S, TIMESTAMP_UNIT_1US

1.6.1.19 Xcp/XcpConfig/XcpEventChannel/XcpEventChannelType

[RTA-BSW] Description	This configuration parameter indicates what kind of DAQ list can be allocated to this event channel. - DAQ indicates only DAQ lists with DIRECTION = DAQ supported - STIM indicates only DAQ lists with DIRECTION = STIM supported
[AUTOSAR] Description	This configuration parameter indicates what kind of DAQ list can be allocated to this event channel.
[RTA-BSW] Enumeration Literals	DAQ, STIM
[AUTOSAR] Enumeration Literals	DAQ, DAQ_STIM, STIM
[REMOVED] Default Value	DAQ

1.6.1.20 Xcp/XcpGeneral/XcpDevErrorDetect

[RTA-BSW] Description	Switches the Development Error Detection and Notification on or off.
[AUTOSAR] Description	Switches the development error detection and notification on or off.
[REMOVED] Introduction	* true: detection and notification is enabled. * false: detection and notification is disabled.
[REMOVED] Default Value	false

1.6.1.21 Xcp/XcpGeneral/XcpMainFunctionPeriod

[RTA-BSW] Description	The XCP does not require this information but the BSW scheduler, which invokes the main function, needs it in order to plan its tasks. Example Period : for 10ms = 0.01
[AUTOSAR] Description	The XCP does not require this information but the BSW scheduler, which invokes the main function, needs it in order to plan its tasks.

1.6.1.22 Xcp/XcpGeneral/XcpNvRamBlockIdRef

[RTA-BSW] Description	This reference is used for Resume Mode to point to the corresponding NvM block for XCP
[AUTOSAR] Description	This reference contains the link to a non-volatile memory block to be used in the feature “RESUME MODE” so this information has to be stored non volatile to be available directly after start-up of the ECU.
[RTA-BSW] Upper Multiplicity	2

[AUTOSAR] Upper Multiplicity	1

1.6.1.23 Xcp/XcpGeneral/XcpTimestampTicks

[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1
[RTA-BSW] Minimum Value	1
[AUTOSAR] Minimum Value	0

1.6.1.24 Xcp/XcpGeneral/XcpTimestampUnit

[RTA-BSW] Description	This parameter indicates the resolution of the data acquisition clock of the slave when transferring data to master. - TIMESTAMP_UNIT_100MS indicates Unit is 100 millisecond - TIMESTAMP_UNIT_100NS indicates Unit is 100 nanosecond - TIMESTAMP_UNIT_100PS indicates Unit is 100 picosecond - TIMESTAMP_UNIT_100US indicates Unit is 100 microsecond - TIMESTAMP_UNIT_10MS indicates Unit is 10 millisecond - TIMESTAMP_UNIT_10NS indicates Unit is 10 nanosecond - TIMESTAMP_UNIT_10PS indicates Unit is 10 picosecond - TIMESTAMP_UNIT_10US indicates Unit is 10 microsecond - TIMESTAMP_UNIT_1MS indicates Unit is 1 millisecond - TIMESTAMP_UNIT_1NS indicates Unit is 1 nanosecond - TIMESTAMP_UNIT_1PS indicates Unit is 1 picosecond - TIMESTAMP_UNIT_1S indicates Unit is 1 second - TIMESTAMP_UNIT_1US indicates Unit is 1 microsecond In product line MDG1, please use 1, 10 or 100 US.
[AUTOSAR] Description	This parameter indicates the resolution of the data acquisition clock of the slave when transferring data to master.
[RTA-BSW] Lower Multiplicity	0
[AUTOSAR] Lower Multiplicity	1
[RTA-BSW] Enumeration Literals	TIMESTAMP_UNIT_1PS, TIMESTAMP_UNIT_10PS, TIMESTAMP_UNIT_100PS, TIMESTAMP_UNIT_1NS, TIMESTAMP_UNIT_10NS, TIMESTAMP_UNIT_100NS, TIMESTAMP_UNIT_1US, TIMESTAMP_UNIT_10US, TIMESTAMP_UNIT_100US, TIMESTAMP_UNIT_1MS, TIMESTAMP_UNIT_10MS, TIMESTAMP_UNIT_100MS, TIMESTAMP_UNIT_1S
[AUTOSAR] Enumeration Literals	TIMESTAMP_UNIT_100MS, TIMESTAMP_UNIT_100NS, TIMESTAMP_UNIT_100PS, TIMESTAMP_UNIT_100US, TIMESTAMP_UNIT_10MS, TIMESTAMP_UNIT_10NS, TIMESTAMP_UNIT_10PS, TIMESTAMP_UNIT_10US, TIMESTAMP_UNIT_1MS, TIMESTAMP_UNIT_1NS, TIMESTAMP_UNIT_1PS, TIMESTAMP_UNIT_1S, TIMESTAMP_UNIT_1US

1.6.1.25 Xcp/XcpGeneral/XcpVersionInfoApi

[REMOVED] Introduction	TRUE: XCP_GetVersionInfo() API service exists FALSE: XCP_GetVersionInfo() API service does not exist
[REMOVED] Default Value	false

1.7 API Definition

Table 1.209.Xcp Supported API functions

`Xcp_Init`	Initializes interfaces and variables of Xcp.
`Xcp_GetVersionInfo`	Returns the version information of this module.
`Xcp_MainFunction`	Scheduled function of the XCP module.
`Xcp_<Lo>RxIndication`	Called by lower layers (FrIf CanIf SoAd or CDD) when an Xcp PDU has been received.
`Xcp_<Lo>TxConfirmation`	Called by lower layers (FrIf CanIf SoAd or CDD) when an Xcp PDU has been transmitted.
`Xcp_<Lo>TriggerTransmit`	Within this API the upper layer (called) module checks whether the available data fits into the buffer size reported by PduInfoPtr→SduLength.

For further information on these functions, please refer to AUTOSAR_SWS_XCP.pdf

1.8 Configuration Advice

NOTE

Before configuring this module, please note that some of the default values used by ConfGen to produce the default ECU Configuration can be changed, and this is done by either adding a Conf-Gen Enhancer Mapping (CEM) or by adding a rule to an *algo.properties* file. For more information, see the "Configuration Generation" section of the RTA-CAR User Guide.

1.8.1 XcpMainFunctionPeriod

XcpMainFunctionPeriod should always be configured using a float value.

1.8.2 Seed and Key

If you want XCP resources to only be accessible when a valid key has been sent by the master, you will need to implement several functions on the XCP slave. Alternatively, you can implement your own algorithm. See the Integration Advice for more information.

If you choose to implement your own algorithm, you'll need to set the SeedAndKey parameter to True (in XcpGeneral) to enable the callout, and set XcpSeedAndKeyExternalFunction (also in XcpGeneral) to the filename that the

XCP master will use to support the seed and key functionality. This file name needs to include the file extension, but not the path. It will be printed in the A2L file for the XCP master to read, but there is no validation from the XCP software.

1.8.3 Page Switching (PAG)

If your ECU supports direct memory access to all calibration values, you do not need to enable Page Switching (PAG) in the configuration of XcpGeneral.

If, however, your ECU supports different memory pages of segments, e.g. reference page and working page, you will not be able to copy every calibration variable via direct MemCopy, and so you need to enable Page Switching in the configuration of XcpGeneral (XcpPageSwitchingPag).

You will then need to integrate several additional functions as described in the Integration Advice.

1.8.4 Static Address Transformation

If the addresses from the A2L cannot be used directly, e.g. for Hook-Based Bypass (HBB), it is possible to enable Static Address Transformation (via the XcpStaticAddressTransformation configuration parameter in XcpGeneral) which will activate the XcpAppl_StaticAddressTransformation callout.

This function is called for each SET_MTA command (UPLOAD/DOWNLOAD) and during the configuration of DAQ/STIM for each OdtEntry. It allows the given address to be transformed into another address (accessed by Xcp) based on the address extension.

It is called Static Address Transformation because the addresses for each OdtEntry are transformed once during configuration, and not for each access, and it can be done with the XcpAppl_OdtEntryRead and XcpAppl_OdtEntryWrite functions. XcpAppl_OdtEntryRead/XcpAppl_OdtEntryWrite can be used when the software is running in a dynamic memory environment where the addresses change from one start to another.

1.8.5 Memory Access

If a simple MemCopy is able to copy data into the ECU and to read data from the ECU, you can switch the XcpUploadDownloadMemoryAccessByAppl configuration parameter (in XcpGeneral) to False.

However, if you want to use a specific function to handle the data transfers instead of a simple MemCopy (e.g. DME), you will need to set XcpUploadDownloadMemoryAccessByAppl to True.

You will also need to integrate several functions, as described in the Integration Advice.

1.8.6 Memory Access for OdtEntries

If a simple assignment is able to copy data into the ECU and to read data from the ECU, you can set the XcpDaqStimMemoryAccessByAppl configuration parameter (in XcpGeneral) to False.

However, if you want to use a specific function to handle the data transfers

instead of a simple assignment, you will need to set XcpDaqStimMemoryAccess-ByAppl to True.

You will also need to integrate several functions, as described in the Integration Advice.

1.8.7 Memory Protection

The Memory Read and Write access protection for DAQ (Command Upload) and STIM (Command Download) is controlled by the XcpMemoryReadProtection and XcpMemoryWriteProtection configuration parameters (in XcpGeneral). See the Integration Advice for details on the application functions.

1.8.8 BYTE ORDER OF CONTROLLER

If you want to measure and calibrate on the Little/Big Endian target, you will need to configure the PlatformRbCpuByteOrder parameter (in the PlatformRbSettings container).

1.9 Integration Advice

1.9.1 Init Functions

Xcp is initialised by Xcp_Init.

At a cold start, the session and DAQ measurement configuration is reset and the state changes from XCP_UNINIT to XCP_INIT. On a subsequent initialization (e.g. caused by a software reset afterwards) this data is left intact, when it is placed within a reset-safe memory section.

1.9.2 DeInit Functions

No specific handling necessary.

1.9.3 Scheduled Functions

Xcp_MainFunction

This function, which is independent of other Xcp functions, should be scheduled in one cyclic task of 10ms or faster.

Xcp_EthMainFunctionTx

This function triggers the transmission of Ethernet frames. It should be scheduled in the fastest cyclic task that is used for measurement (call to Xcp_EventChannel). If that task is not cyclic (e.g. engine sync task), this function should be called in a cyclic task that is always faster than the non-cyclic task. In order to have the lowest delay, this function should be scheduled after Xcp_EventChannel.

Xcp_FrMainFunctionTx

This function triggers the transmission of FlexRay frames, and it should be scheduled once in a FlexRay cycle. It should be called in the same task/interrupt context as FrIf_JobListExec_0, and these two functions must not interrupt each other. For the lowest response delay, Xcp_FrMainFunctionTx should be scheduled

directly after the reception of XCP frames and before the transmission of the next XCP frames. This can be done with the help of the FrIfJobListCallback parameter (inside the FrIfJobListWizard configuration of FrIf).

Xcp_EventChannel

This function, which is independent of other Xcp functions, is used to sample measurement data. It should be scheduled in all cyclic tasks where XCP measurement is to be done. For each configured XcpEventChannel (e.g. Xcp_EventChannel_10ms) generated macros rather than Xcp_EventChannel should be called, but it can optionally be called with any valid XcpEventChannelNumber as a parameter.

If the user enables the boolean parameter XcpEventChannelScheduling, all the Xcp_EventChannels will be schedulable entities. In this case, both the schedulable entities and module entry information will be generated in the Xcp_Cfg_BSWMD.arxml file after a successful build.

If the XcpEventChannelScheduling parameter is disabled (which is the default), the event channel information is generated as called entities and it needs to be called in another function.

To avoid double declaration-related warnings in cases when RTE is used, it is ensured that the definitions for schedulable entities are not available in *Xcp_Cfg.h* as they are made available via the generated file *SchM_Xcp.h*. In cases where RTE is not in use, the EventChannel declarations are generated in the *Xcp_Cfg.h* file.

- The call can be placed in a task or in any function.
- It is not mandatory to call cyclically (but you must ensure that the cycletime is configured accordingly).
- For data acquisition the call should be placed right after where all data that shall be measured is available.
- For data stimulation the call should be placed right before where the data is actually used.

1.9.3.1 Post Build

Xcp does not itself support postbuild, which means that Xcp PDUs (transmit or receive) cannot be configured in the Xcp configuration to be available in only some Postbuild variants. PDUs that are used by the Xcp must be available in **all** configured postbuild variants.

1.9.3.2 External Checksum

Xcp doesn't support a checksum calculation by itself, so if you want to use a checksum calculation (e.g. over a memory segment or a calibration page) you'll need to integrate the XcpAppl_BuildChecksumTrigger and XcpAppl_BuildChecksumMainFunction functions.

You can use the Xcp_BuildChecksum.c file to implement your algorithm.

1.9.3.3 Seed and Key

If you want XCP resources to only be accessible when a valid key has been sent by the master, you will need to integrate the XcpAppl_GetSeed and XcpAppl_Unlock functions on the XCP slave.

You can optionally rework the Xcp_SeedAndKey.c file to implement your own algorithm. You must then enable the callout for your own seed and key application functions, as described in the Configuration Advice.

For the XCP master, a seed and key functionality in the form of a library must also be provided.

1.9.3.4 Page Switching (PAG)

If your ECU supports different memory pages of segments, e.g. reference page and working page, you will not be able to copy every calibration variable via direct MemCopy, and so you need to enable Page Switching in the configuration (see Configuration Advice) and integrate the following functions:

- XcpAppl_SetCalpage
- XcpAppl_GetCalPage
- XcpAppl_CopyCalPage
- XcpAppl_GetPagProcessorInfo
- XcpAppl_GetSegmentInfo
- XcpAppl_GetPageInfo

The Xcp_CalPage.c file contains example stubs for these functions.

1.9.3.5 Memory Access

If you want to use a specific function to handle the data transfers instead of a simple MemCopy (e.g. DME), you will need to integrate the following functions:

- XcpAppl_MemRead
- XcpAppl_MemWrite
- XcpAppl_MemWriteMainFunction

The Xcp_MemoryReadWrite.c file contains example stubs for these functions.

1.9.3.6 Memory Access for OdtEntries

If you want to use a specific function to handle the data transfers instead of a simple assignment, you will need to integrate the following functions:

- XcpAppl_OdtEntryRead
- XcpAppl_OdtEntryWrite

The file Xcp_OdtEntryReadWrite.c provides sample stubs for these functions.

1.9.3.7 Memory Protection

Memory Read and Write access protection is available for DAQ (Command Upload) and STIM (Command Download) - see the Configuration Advice for more details on how to enable these features.

Application functions then decide whether the access to a requested memory area is granted or denied. The Xcp_MemoryAccessProtection.c file provides sample stubs for these functions.

By default, Xcp allows read access to GetID memory even if the XcpAppl_MemReadProtection function denies it. Similarly, by default, Xcp write access is enabled for Xcp critical information area (i.e Xcp_Instance) even if the XcpAppl_MemWriteProtection function denies write access.

1.9.4 OS Resources

The number of the active OS resources, and their usage, depends on the user configuration of the AUTOSAR System and Basic Software. It is the user's responsibility to ensure that the `ENTER` and `EXIT` resource API are implemented to disable/enable interrupts as required. Alternatively they must be configured in the OS with the correct task/ISR allocation.

The following is a list of all exclusive areas that this module might use regardless of the user configuration. It is the job of the integrator to ensure that the implementation of these exclusive areas is correctly defined. This implementation can be done using the RTE exclusive area configuration wizard or by manual implementation of the corresponding BSW API calls to ensure exclusivity.

The Xcp module uses the following OS resources:

- Xcp_SendingShort
- Xcp_DownloadNoNest
- Xcp_BufferAccessActiveNoNest
- Xcp_EthMainFunc
- Xcp_CanBuffering
- Xcp_FrTransmit
- Xcp_GetTimestamp
- Xcp_DaqRam

1.10 Hardware Requirements

Table 1.210.Xcp Hardware Requirements

Support	Usage
FLASH	11595 bytes
RAM	6912 bytes

Usage calculated based on a sample configuration.

2 Glossary

A2L
File Extension for an ASAM MCD-2 MC Language File

ASAM
Association for Standardization of Automation and Measuring Systems

CAN-FD
An extension to the original CAN bus protocol

CanIf
AUTOSAR CAN Interface module

DAQ
Data AcQuisition

DTO
Data Transfer Object

HBB
Hook Based Bypass

MCD-2 MC
ECU Measurement and Calibration Data Exchange Format

ODT
Object Descriptor Table

SoAd
AUTOSAR Socket Adaptor module

STIM
Data STIMulation packet

XCP
Universal Calibration Protocol

Xcp
AUTOSAR XCP module